

# City Air Makes You Free

(in some countries or if you are upper class or financially satisfied)

Tuesday 29<sup>th</sup> September, 2026 10:07

**target jou: cities if8.2, ariq if3ish, some soc j have soc lit like toennies simmel maybe even durkheim maybe sth like j or urb soc**

We use World Values Survey 7 (2017-2023) across 64 countries to test the hypothesis that cities are more free. Mostly we find the opposite: in 53 out of 64 countries there is mostly no relationship or even in 6 cases positive relationship. Still, in 16 countries cities are indeed more free. Another key finding is that, as already theorized by Toennies, the upper classes (and financially satisfied) are “active and alive” in cities or as we find here autonomous/free. And the poor, lower class (and financially dissatisfied) are less free/autonomous in cities. Within-country urbanicity freedom/autonomy differences’ effect sizes tend to be substantial, about 0.3-0.6 and as large as around 1 on 1-10 autonomy-freedom scale. The whole sample pooled effect sizes of financial satisfaction and class interactions with urbanicity are substantial of about 2 steps on 1-10 income scale.

WORLD VALUES SURVEY, FREEDOM, AUTONOMY, URBANISM, URBAN, RURAL, URBAN-RURAL, CITIES

City life and Gesellschaft down the common people to decay and death. Ferdinand Toennies

In the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive. Ferdinand Toennies

City air makes you free<sup>1</sup>—the saying originated in the Middle Ages, and it meant freedom from feudalism in cities, non-feudal islands in a sea of feudalism (Harvey 2012).

The parallel is somewhat a stretch—and the title is somewhat a hook bait. The original saying is from the Middle Ages and meant freedom from feudalism in cities. Feudalism long has been gone, but we may actually be living under feudalism again, so called techno-feudalism (Varoufakis 2024, Morozov 2022). While there is no reason to believe that modern cities are more free from techno-feudalism, as Middle Ages cities used to be more free from feudalism, the narrative remains that cities are still more free than ruralites, especially for non-conformist types such as LGBTQ.

freedom/autonomy is a good thing in just about any respect except perhaps responsibility—for instance in the middle ages there was little freedom, but also little responsibility—for instance the poor and the slaves were not responsible for their lot—it was God given. Now the poor have to worry about being poor and about it being their fault—everyone is free after all and their achievements and experiences are of their own making. Still, blaming the poor for their lot is sometimes higher in less free countries such as the US than in more free countries such as Mexico or Colombia.

It’d be already a finding worth reporting that people don’t have more freedom/autonomy in cities, because there are theories, reasons, and arguments that they do. But we also have the case why there would be less freedom/autonomy in the city, just as there would be less in the West (especially the US) v Latin America. Mexico, Colombia and Venezuela are some of the most free countries in the World (Okulicz-Kozaryn 2014, Okulicz-Kozaryn and Martinez 2025).

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<sup>1</sup>(Stadt Luft macht frei) (Park et al. [1925] 1984, p. 12)

It is counterintuitive. City, just as the US appears the land of the free, but it is not. Freedom/autonomy can be counterintuitive—for instance the US advertises itself as “the land of the free” and yet people in the US feel less free at 7.7 than Mexicans and Colombians at above 8.1. Likewise, city may seem more free than ruralites, but as this paper argues, usually that’s not the case.

City (and the US) is stressed, hectic, anxious, materialistic, consumerist (Okulicz-Kozaryn 2015), and overstimulating (Simmel 1903, Lederbogen et al. 2011)—in such conditions freedom/autonomy is difficult. In a city v rural (or the US v Latin America) there is more social pressure,<sup>2</sup> and so an urbanite is more of an “automaton” (to use Fromm’s term) (Okulicz-Kozaryn 2014).

## 1 Freedom/autonomy defined

Some authors use term “freedom” (Okulicz-Kozaryn 2014) and some “autonomy” (Steckermeier 2022)—we use the terms interchangeably.

What is freedom/autonomy?<sup>3</sup> To be free is to be able to choose, to control and direct one’s own life. Freedom is having control—a person is free is she is her own master. It could be labeled “psychological freedom:” “Freedom means a degree of harmony between basic motives and overt behavior” (Bay 1970, p.83 cited in Okulicz-Kozaryn 2014). It is “capability to choose,” “awareness of alternatives” and “courage to choose” (Veenhoven 2000).<sup>4</sup> Clearly, there are more choices in city v ruralites, so at least in a sense of choice there is more freedom. But then too much choice may also backfire and even lead to a paralysis (Schwartz 2004).

Freedom/autonomy is a broad concept and could be sub-divided into dimensions—Hitlin and Long (2009) reviews the concept in sociology. See also a recent scale (D’Italia and Okulicz-Kozaryn 2025). Another useful review is Kouba (2016) and also relevant here as it points to our WVS measure.

## 2 Literature

**leonie help! im a geographer not a sociologist; Leonie has comprehensive 2yr old lit rev on predictors of autonomy and will know if anybody used urb rur even only as control**

There are theoretical reasons to expect more freedom in cities v ruralities. In rurality there is mechanical solidarity—rural folks function like parts of a simple machine—groups are held together through homogeneity and similarity: similar lifestyles, roles, shared values and morality (Durkheim [1893] 1997). Hence, ruralities force social conformity not conducive for individual freedom. On the other hand, organic solidarity in cities allows for more differentiation and individualism. Urbanites are also more tolerant (Tuch 1987). Likewise, there would be more *Gemeinschaft* in ruralities—more collective and social that may restrict personal freedom; and more *Gesellschaft* in cities that may promote individualism (but also hostility and aggression). Simmel observed that as urbanism overloads a person it forces to distance oneself from the environment (Simmel 1903), which also may be conducive for differentiation, individualism, and then perhaps freedom. But urban hunkering down and blasé attitude is not agentic.

Many studies approach freedom as predictor of subjective wellbeing, but freedom is also an end-in-itself (Okulicz-Kozaryn 2014), as recognized by Fromm (2013, 2012, 1992, 1964, 1944, [1941] 1994) and Sen (2000). The Universal Declaration of Human Rights states that freedom is a human right: freedom for the realization of one’s human potential (art. 26, 29).

There are studies on freedom using WVS variable that we use here (e.g., Okulicz-Kozaryn 2014, Pitlik 2016, Nikolaev and Bennett 2016), but they do not examine urbanicity. There are many studies on the effect of urban-rural on subjective wellbeing (SWB) (e.g., Sørensen 2024, 2014, Knies 2026, Okulicz-Kozaryn and Valente 2021), but the effect of urban-rural on freedom has only been studied once. The only such paper is Yamagishi et al. (2012), including among the papers that cite it.

<sup>2</sup>Also see a fantastic book by Frank (2012) on keeping up with the Joneses in the US.

<sup>3</sup>Here we focus on personal as opposed to societal freedom and follow Okulicz-Kozaryn (2014) in this paragraph.

<sup>4</sup>The degree of perceived control over choice is called “locus of control” in psychology and Verme (2009) argues that WVS item used here measures locus of control in addition to freedom.

Yamagishi et al. (2012) tests our “city free air” hypothesis: the social constraints prevalent in rural life are weaker in metropolitan areas, freeing metropolitan residents from pressure to suppress their pursuit of individual goals. Yet Yamagishi et al. (2012) test is very different from ours—they show that urbanites are more likely to chose extravagant pens than rural folks. Still, the study is useful in at least 2 ways. First they show that indeed there is more freedom in city (using extravagant pens test), and they provide theory and literature summarized in the next paragraph.

Ester and Vinken (1993) is not about urban-rural freedom specifically, but it is indirectly related. Ester and Vinken (1993) uses similar dataset to ours, EVS/WVS, and tests yuppie value syndrome (yuppies are young urban professionals), and finds that yuppies tend to hold liberal social and ethical views, but are not as hypothesized: economically conservative, absent from politics, and don’t have distinctive lifestyle emphasis.

Cities provide freedom from traditional rural social constraints—Gesellschaft v Gemeinschaft—deviation from community norms is more likely to be reprimanded and ostracized in rural v urban. An urbanite may be more internally driven rather than externally constrained rural folk. City provides more opportunities for alternative jobs and social relations, so urbanite is more free by being less concerned about social exclusion as she has plenty of alternatives. Also see discussion in Luca et al. (2023).

One mechanism is higher social pressure for conformity in ruralities—cities provide anonymity—indeed concern for moral reputation is greater in rural than urban areas (Danielson et al 2023 cited in Okulicz-Kozaryn 2026). The higher the population density, the lower the frequency of social comparison measuring social comparison with searches on Google (Okulicz-Kozaryn 2026).

But there is also theory suggesting less freedom in cities. For instance Steve Pile argues that old cities can be a weight on its inhabitants with their history and past (Pile 2005a,b, Pile et al. 1999), and old cities carry melancholia (Pile 2005b). All of that can translate into less freedom.

### 3 Hypotheses

$H_1$  : People in cities are freer than people in rural areas.

???? i would either use the full scale (pseudo-metric) or make a median split by country here

yeah its a mix some countries this or that, but most sig c yes more free in cities

$H_2$  : This difference is more pronounced in less developed countries

why we hypothesize this?

not much there neither iwth gdp nore inc group

but yes with perc urban—more urban more c, more freedom in city: in fig 10

$H_3$  : The “more” of freedom in cities mostly stems from rich people/upper classes (following the Toennies quote)

### 4 Data and model

World Values Survey (WVS) is a fantastic and free dataset available from <https://www.worldvaluessurvey.org>. It is a great source of data as compared to \$30k+ closed source Gallup data. It is helpful to use open source public data amidst corporatization of research and academia (Mills 2012a, Cox 2013, Mills 2012b, Catropa and Andrews 2020, Schmidlin 2015). WVS is representative of each country. Countries in the sample and sample sizes are shown in appendix.

Autonomy/freedom variable reads: “Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out.”

Validity and reliability of the measure seem fine and the variable has been used successfully in the literature (e.g., Verme 2009, Inglehart et al. 2008, Veenhoven 2000, Okulicz-Kozaryn 2014) but still the question may be interpreted slightly differently in different languages/cultures, and some caution would be useful. This is also one reason for within-country analyses that we start with in the next section.

Urban-rural is a gradient, not a dichotomy (Okulicz-Kozaryn 2017), but here for simplicity we use 3 or 4 collapsed categories and focus on the endpoints of the spectrum:  $> 500k$  v small  $< 10k$  or  $< 20k$  (separately for each country).  $> 500k$  represents city—indeed it could even be bigger than 500k—urban unhappiness starts around several hundred thousand (Okulicz-Kozaryn 2017).

We dropped countries where there were fewer than 50 observations in cities  $> 100k$ . And in appendix as a robustness/sensitivity check with fewer than 50 observations in cities  $> 500k$  *iii*

All variables including controls are listed in appendix and there are also shown distributions of all the variables. We are controlling for age, age<sup>2</sup>, male, married, employment type dummies (full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and importance of religion.

Per controls we follow Pitlik (2016) except that we use financial satisfaction in most specifications instead of income as it arguably predicts freedom more than income and indeed in our sample correlates higher, and also we don't use their country level democracy index, and use PCGDP in appendix only. Likewise, we have all person level predictors that Nikolaev and Bennett (2016) have used.

Money is important for freedom. Indeed as many financial gurus on YouTube would tell you freedom is the only or main reason to accumulate wealth. For instance, the FIRE movement (Financial Independence, Retire Early) is all about freedom/autonomy. Hence in our analyses we will focus on income, class and financial satisfaction. Financial satisfaction is the key—not everyone needs the same amount of money to be financially satisfied and feel free. It is measured as “How satisfied are you with the financial situation of your household?”

We were tempted to add trust, crime victimization and fear of crime, as these correlate with urbanicity and may predict freedom. But eventually we do not include them because those are rather caused by urbanicity, not the other way round, and hence may lead to overcontrol bias:

What we really need, to identify/estimate  $X \rightarrow Y$ , is to control for other determinants of  $Y$  that are also antecedents of  $X$  (so,  $W \rightarrow X$ ). If we include controls where the relationship goes the other way ( $X \rightarrow W$ ), we will exacerbate bias in our estimate of  $X \rightarrow Y$ . (The relevant term for that situation is 'overcontrol bias') (Bartram et al. 2024, p. 5).

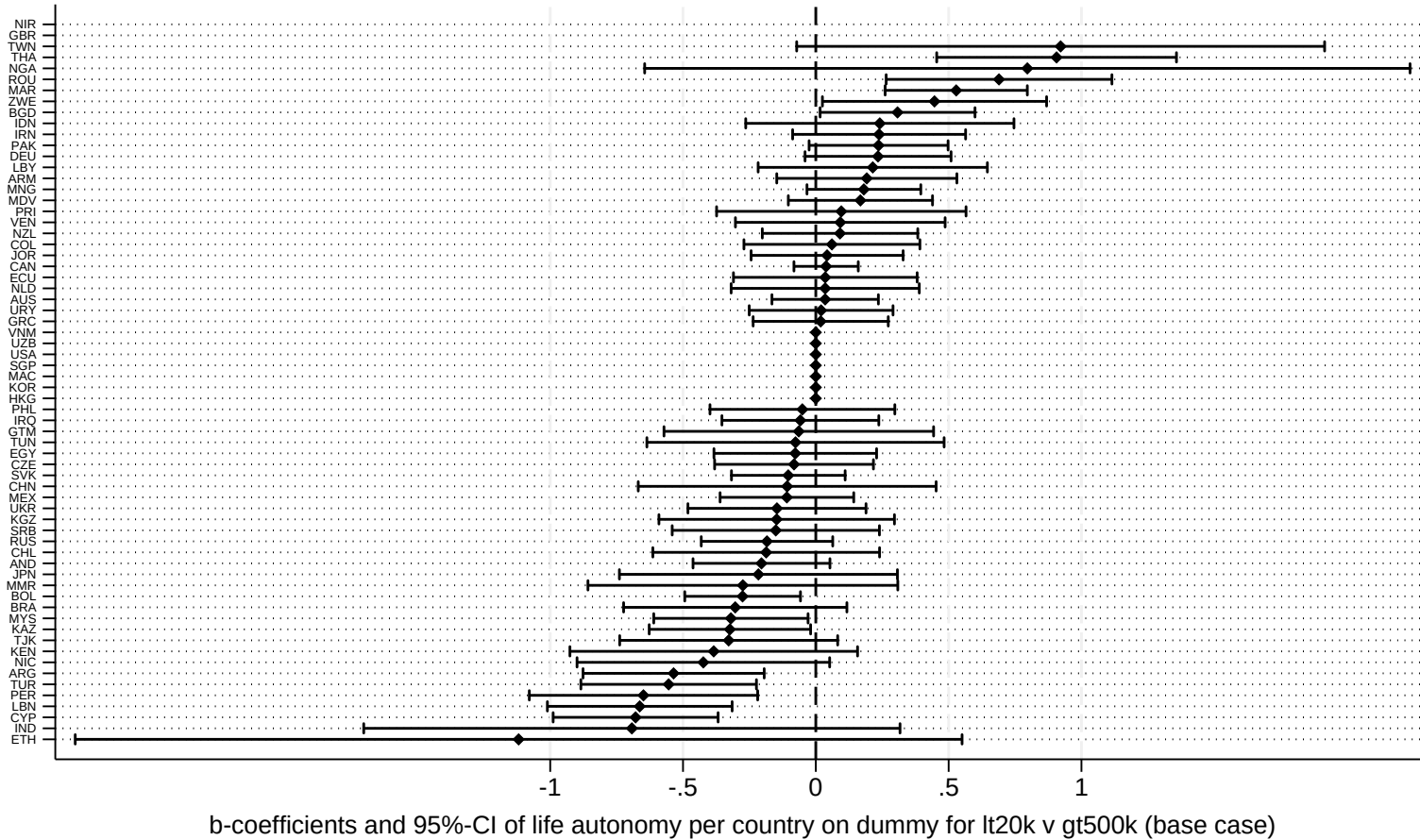
## 5 Results

A natural first approach would be to find overall pattern on pooled complete dataset across all countries to see if there are universal patterns or at least by income group or region. Indeed this is how we have started, but we postpone these results to the appendix as we have learned through our sensitivity/robustness checks that inclusion of country dummies yields all urban-rural results insignificant. Hence we simply start with the results by country.

Financial satisfaction is the strongest correlate of freedom (0.33), stronger than health (0.21), class (0.14), and income (0.09). Financial satisfaction is also related to class (0.26) and income (0.34).

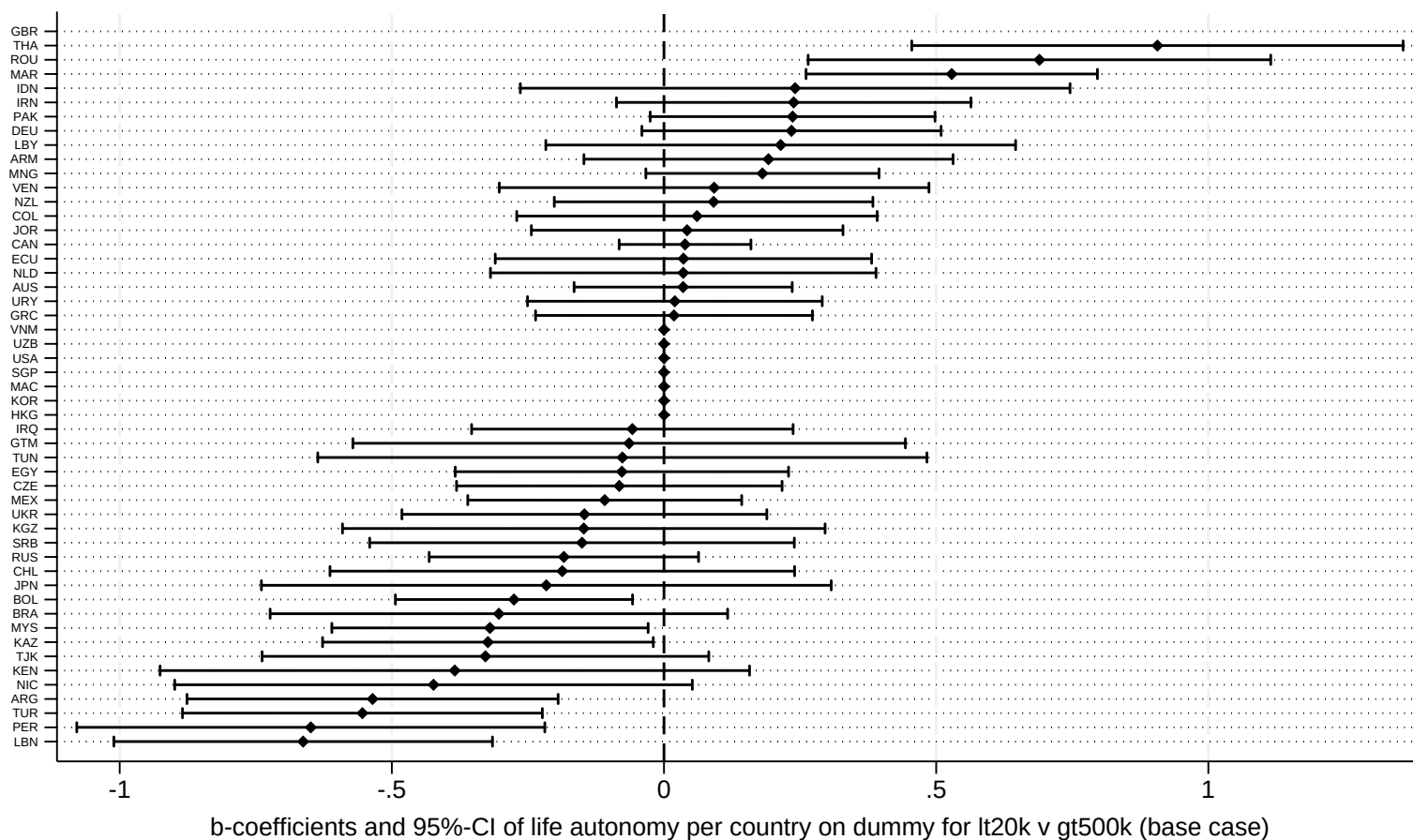
## 5.1 By country

Results are in figure 1. Is there less freedom in smaller areas as often argued? Are cities more free? No. Most countries show insignificant urban-rural differences at  $< 0.05$  level of significance. Again, it is already a finding worth reporting that people don't have more freedom/autonomy in cities, because there are theories, reasons, and arguments that they do.

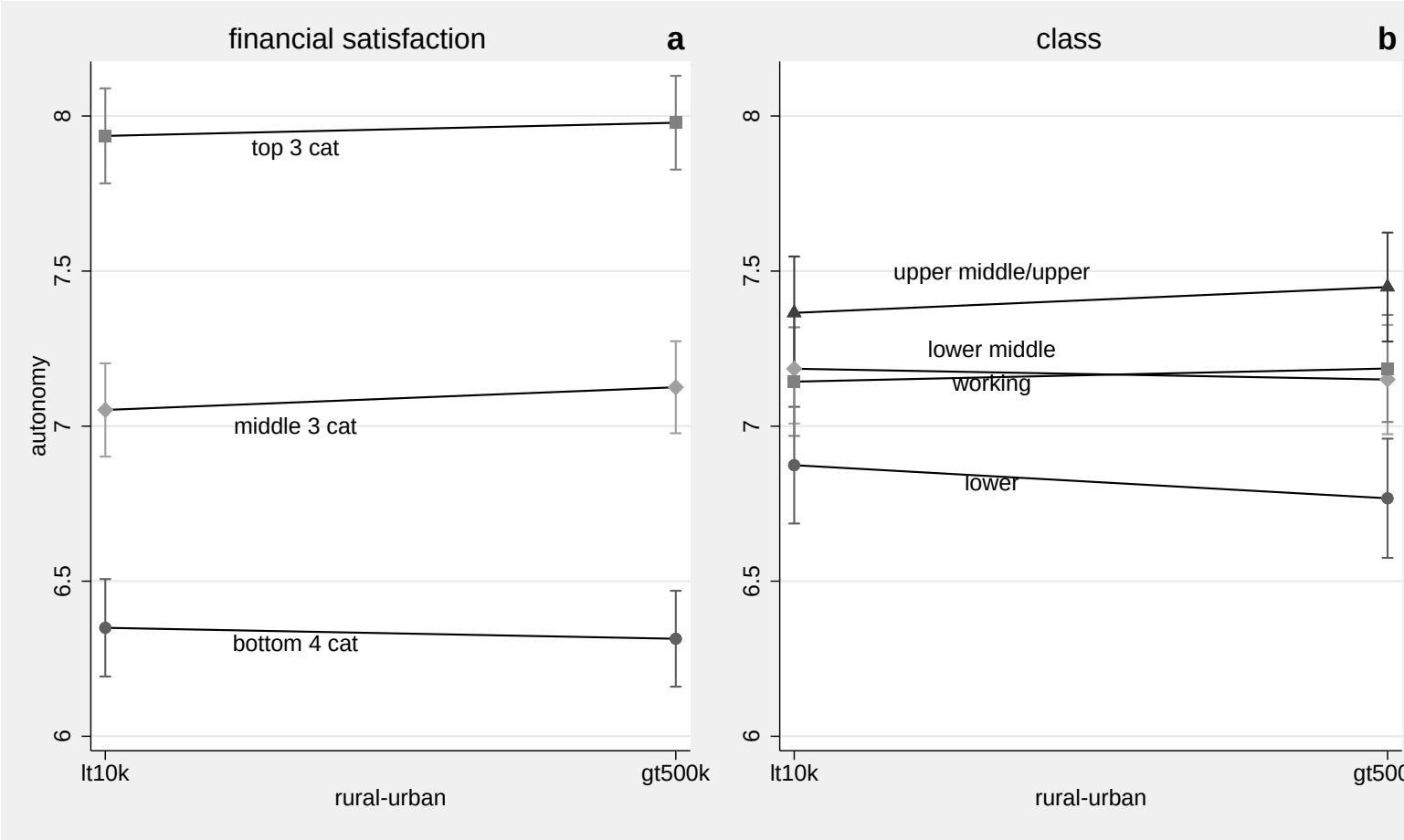


**Figure 1:** Using model controlling for age, age<sup>2</sup>, male, married, employment type dummies(full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and religion important. See appendix for results in table also including omitted here 20 – 500k category. See appendix model for pooled sample in table 3 except that original 8 step urbanicity variable is now categorized into 3 bins as there are smaller cell sizes separately by country. Still note that few countries lack the estimates as even this simplified 3 step urbanicity spectrum is not available. Overall the results are mostly insignificant, but more negative (11) than positive (6). So in general within countries for few significant relationships there is less freedom in smaller areas as theorized in the literature.

Only 16 out of 64 countries have more freedom in cities. They have at least one coefficient on either  $< 20k$  or  $20 - 500k$  ( $> 500k$ ) negative and significant at  $< 0.05$ . 6 countries, on the other hand, have more freedom in ruralities—they have positive effect coefficients on smaller areas dummies. Note that figure 1 only shows  $< 20k$  coefficients; for  $20 - 500k$  coefficients see appendix.



**Figure 2:** cc goes down from 66 to 51 when dropped if gt500k missing; same pattern as earlier most insignificant, some positive but most negative



**Figure 3:** on the sample after dropping countries where top ght500k cat missing

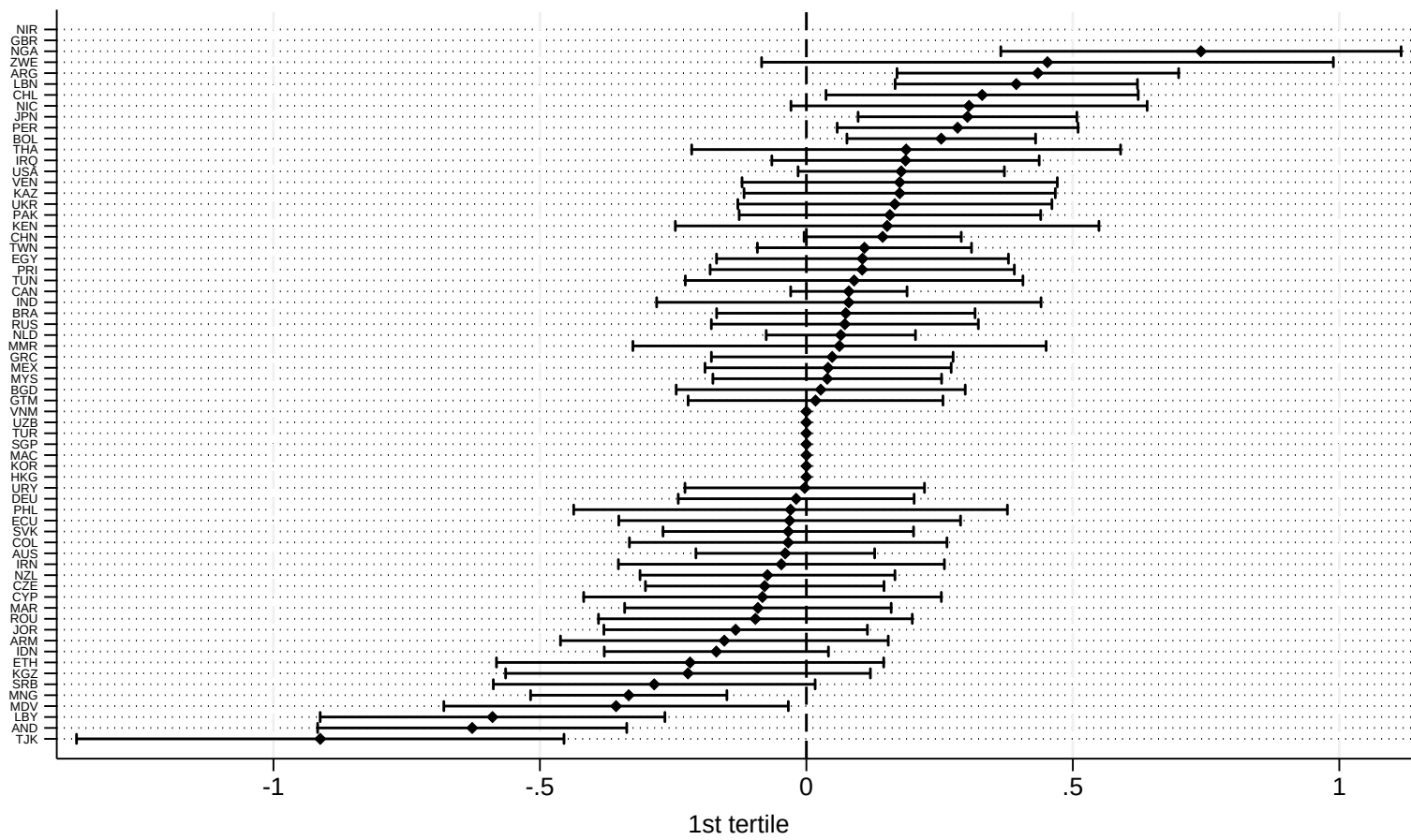
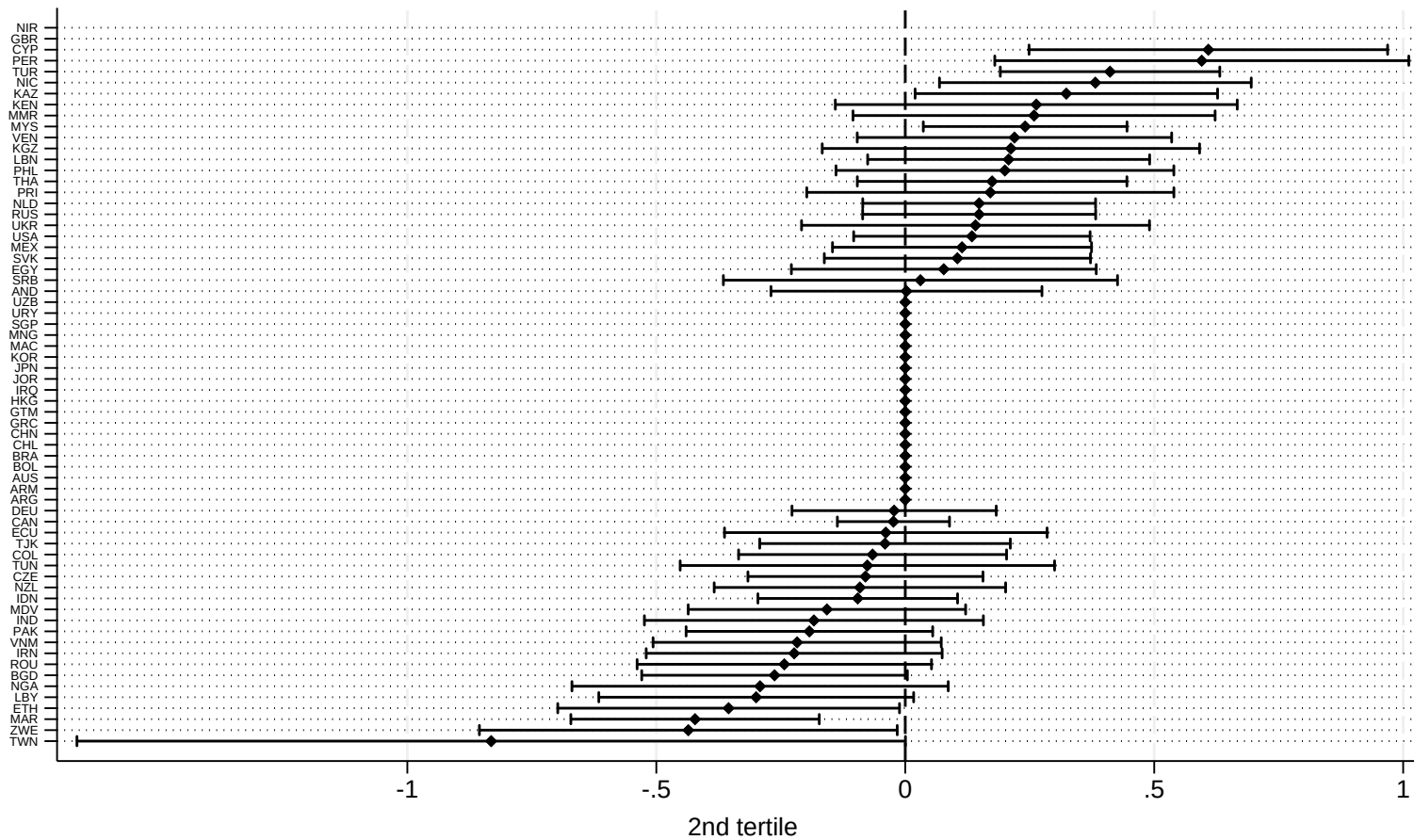


Figure 4: 2nd tertile





**Figure 5:** 3rd tertile

It is difficult to see any consistent country patterns in the table. Perhaps what stands out is that among the richer or more economically developed countries there are smaller differences. In poorer/less economically developed countries there are more differences: mostly less freedom in smaller places.<sup>5</sup> This makes sense—smaller places in developing countries can be very traditional and very poor—freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, no need for a large city. This is similar to what happiness literature have found with respect to happiness (Okulicz-Kozaryn 2015)—only in the very poorest places of all such as Sub-Saharan Africa happiness is greater in larger cities as life is unbearable and lacking necessities such as shelter and clean water in ruralities.

Countries could be classified by region (see appendix). About half of Latin countries in this sample have lower freedom in smaller areas: ARG, BOL, CHL, NIC, and PER; and about the other half are insignificant: BRA, COL, ECU, GTM, PRI, URY, and VEN.

South Asia does not have a clear pattern: BGD is positive and PAK marginally positive, but IND highly negative, and MDV is insignificant.

Europe/Central Asia: mostly insignificant: ARM, CZE, GRC, KGZ, NLD, RUS, SRB, SVK, and UKR. But otherwise mostly negative: AND marginally, CYP, KAZ, TJK marginally, and TUR; and positive: ROU, DEU marginally. Note that GBR has missing obs and does not appear in the table at all, and UZB has missing urbanicity data.

East Asia/Pacific is mostly insignificant: AUS, CHN, IDN, MMR, NZL, PHL. And several countries in East Asia/Pacific are without respondents in smaller areas: HKG, KOR, MAC, SGP, and VNM. Positive: MNG, and highly positive THA; and negative: JPN, MYS.

Middle East/North Africa mostly insignificant: EGY, IRN, JOR, LBY, and TUN. Two negative: IRQ, LBN; and one positive: MAR.

<sup>5</sup>But sometimes more.

Sub-Saharan Africa: two insignificant: KEN, NGA; ETH is negative, but ZWE positive.

Two countries missing the region classification by the World Bank: NIR and TWN. NIR has a small sample size of less than 500 and missing observations on the variables used in regression so it does not appear in the table. TWN is insignificant.

## 5.2 Interactions with financial satisfaction, class, and income

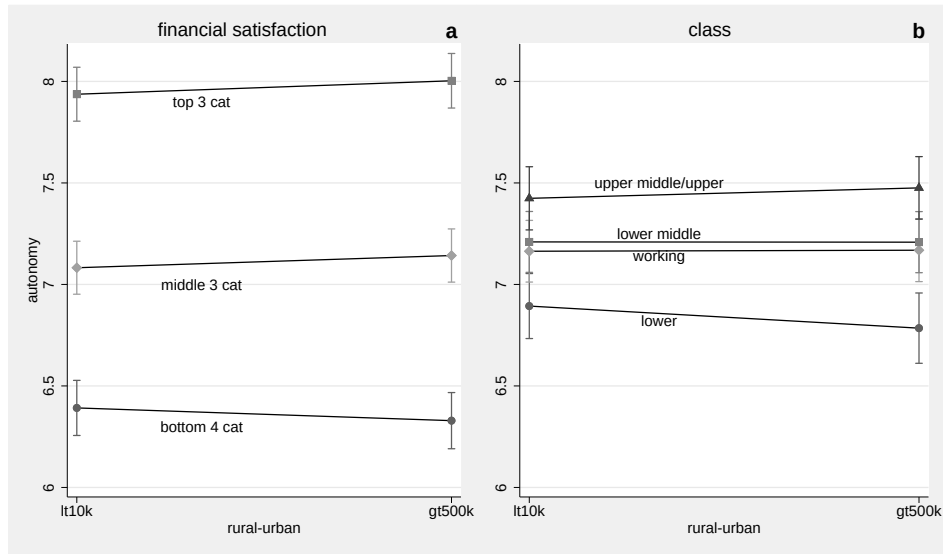
Again the initial results with pooled sample for all the countries (in appendix) turned out to be all insignificant when including country dummies, hence we started with results within each country in the previous table.

But there is one significant result for the pooled sample on urbanicity when controlling for countries—when urbanicity is interacted with financial satisfaction, class, and income. While earlier we just used OLS within each country, here we use multilevel modeling (MLM) with random intercept at country level.

Results are visualized in figure 6.<sup>6</sup>

In panel a by financial satisfaction those in top 3 categories are at about 8 v middle 3 at slightly above 7 v bottom 4 at below 6.5. For the bottom categories there is very slight decrease from rural to urban, and for the middle and upper financial satisfaction categories there is a slight increase from rural to urban. In panel b class: Upper middle and upper classes are more free in largest cities, while lower class is more free in smallest places. In other words, freedom differences are magnified in largest cities for both financial satisfaction and class—the financially satisfied and higher classes enjoy higher freedom premium over financially dissatisfied and lower class in cities.

Income in panel c agrees for the bottom 2 quintiles where freedom declines from  $< 10k$  to  $> 500k$ , and opposite pattern for 3rd middle quintile and 4th quintile where freedom slightly increases from  $< 10k$  to  $> 500k$ . But the top quintile, counterintuitively, behaves like the bottom 2, and we do not have an explanation for this.



**Figure 6:** Multilevel country random intercept regressions of freedom on rural-urban dummies interacted with: a) financial satisfaction, and b) class. For underlying regressions see appendix. Controls are the same as in the previous section.

It could be interesting to separate these results by country, however we leave it for the future research as sample sizes run small

<sup>6</sup>We focus here on visual interpretations using marginsplot, but underlying regressions are in appendix. The relevant significant interactions are for smallest places  $< 10k$  (against the base case of  $> 500k$  with top category(ies) for financial satisfaction in first column a for middle financial satisfaction (v bottom base case) at  $-0.12$  and the top category at  $-0.13$ . Likewise, in column b for class interactions, there is significant interaction of  $< 10k * Uppmid/upper$  at  $-0.16$ . Thus, folks of both high financial satisfaction and class are less free in smallest areas v the largest areas. In last column c with income, there are no significant interactions except unexpected top income quintile, but other insignificant patterns are in line with financial satisfaction and class: the two bottom deciles are less free in the largest places.

for interactions. But we do it by region and income group for financial satisfaction only and postpone these results to appendix<sup>7</sup>

## 6 Discussion and Conclusion

We find support for Toennies' idea: "in the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive" (Tönnies [1887] 2002, p. 227). There are significant interactions in the pooled sample for financial satisfaction and class.

In the pooled sample without interactions there are no significant urban rural differences when controlling for country dummies. But there are some significant results separately by country. The hypothesis that cities are more free does not find support except in 16 countries out of 64.

It is worth highlighting that finding no effect in most countries is a finding worth reporting—the expectation from theory and literature is that the bigger places have more freedom. And in 6 countries we find a positive effect.

Like in Okulicz-Kozaryn (2024) urbanites probably actually have in a way fuller and more free life, but perhaps they realize more that they miss out relatively to other urbanites, and so they feel relatively less free. One mechanism could be that people tend to compare upwards (Frey and Stutzer 2002) and there are more people to compare to in urban areas, inequality ladders are taller and the pond is bigger—small fish in big pond (city) seem to be less free than big fish in a small pond (rural area).

In poorer/less economically developed countries there are more differences in freedom across rural-urban, mostly less freedom in smaller places<sup>8</sup>—this makes sense—smaller places in developing countries can be very traditional and very poor—freedom is difficult in such places. In richer countries, on the other hand, one can already be free just about anywhere, no need for a large city. This is a similar finding as in literature with respect to happiness (Okulicz-Kozaryn 2015)—only in the very poorest places of all such as Sub-Saharan Africa, happiness is greater in larger cities as life is unbearable and lacking necessities such as shelter and clean water in ruralities. Everywhere else urbanites tend to be less happy (Okulicz-Kozaryn and Valente 2021).

# ONLINE APPENDIX (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead as SOM 'Supplementary Online Material']

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<sup>7</sup>Here just real quick: Financial satisfaction panel a results are stronger in East Asia/Pacific and Sub-Saharan Africa (due to sharp increase for the top tertile), and show similar pattern in Europe/Central Asia except that the middle tertile for financial satisfaction is stronger than the top. Latin America/Caribbean and North America shows the opposite pattern where the bottom tertile has the steepest slope, and in Middle East/North Africa middle and top tertiles even reverse to negative. in South Asia the slopes are parallel—no differences there. By income group: High income is similar to the overall pattern.  
<sup>8</sup>(but sometimes more)

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# 1 Additional info

## 1.1 Country sample sizes; Regional country classification; Variables' definitions (survey questions); Histograms

We use World Bank regions classification into several regions: East Asia/Pacific, Europe/Central Asia, Latin America/Caribbean, Middle East/North Africa, North America, South Asia, and Sub-Saharan Africa. Countries in each region and sample sizes are in appendix.

|                                         |  |        |         |        |
|-----------------------------------------|--|--------|---------|--------|
| . bys region: ta cc                     |  |        |         |        |
| -----                                   |  |        |         |        |
| -> regionname =                         |  |        |         |        |
| ISO 3166-1                              |  |        |         |        |
| alpha-3                                 |  |        |         |        |
| country                                 |  |        |         |        |
| code                                    |  | Freq.  | Percent | Cum.   |
| -----                                   |  |        |         |        |
| NIR                                     |  | 447    | 26.77   | 26.77  |
| TWN                                     |  | 1,223  | 73.23   | 100.00 |
| -----                                   |  |        |         |        |
| Total                                   |  | 1,670  | 100.00  |        |
| -----                                   |  |        |         |        |
| -> regionname = East Asia/Pacific       |  |        |         |        |
| ISO 3166-1                              |  |        |         |        |
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| country                                 |  |        |         |        |
| code                                    |  | Freq.  | Percent | Cum.   |
| -----                                   |  |        |         |        |
| AUS                                     |  | 1,813  | 7.29    | 7.29   |
| CHN                                     |  | 3,036  | 12.21   | 19.50  |
| HKG                                     |  | 2,075  | 8.35    | 27.85  |
| IDN                                     |  | 3,200  | 12.87   | 40.72  |
| JPN                                     |  | 1,353  | 5.44    | 46.16  |
| KOR                                     |  | 1,245  | 5.01    | 51.16  |
| MAC                                     |  | 1,023  | 4.11    | 55.28  |
| MMR                                     |  | 1,200  | 4.83    | 60.10  |
| MNG                                     |  | 1,638  | 6.59    | 66.69  |
| MYS                                     |  | 1,313  | 5.28    | 71.97  |
| NZL                                     |  | 1,057  | 4.25    | 76.22  |
| PHL                                     |  | 1,200  | 4.83    | 81.05  |
| SGP                                     |  | 2,012  | 8.09    | 89.14  |
| THA                                     |  | 1,500  | 6.03    | 95.17  |
| VNM                                     |  | 1,200  | 4.83    | 100.00 |
| -----                                   |  |        |         |        |
| Total                                   |  | 24,865 | 100.00  |        |
| -----                                   |  |        |         |        |
| -> regionname = Europe/Central Asia     |  |        |         |        |
| ISO 3166-1                              |  |        |         |        |
| alpha-3                                 |  |        |         |        |
| country                                 |  |        |         |        |
| code                                    |  | Freq.  | Percent | Cum.   |
| -----                                   |  |        |         |        |
| AND                                     |  | 1,004  | 3.88    | 3.88   |
| ARM                                     |  | 1,223  | 4.73    | 8.61   |
| CYP                                     |  | 1,000  | 3.87    | 12.48  |
| CZE                                     |  | 1,200  | 4.64    | 17.12  |
| DEU                                     |  | 1,528  | 5.91    | 23.03  |
| GBR                                     |  | 2,609  | 10.09   | 33.13  |
| GRC                                     |  | 1,200  | 4.64    | 37.77  |
| KAZ                                     |  | 1,276  | 4.94    | 42.70  |
| KGZ                                     |  | 1,200  | 4.64    | 47.35  |
| NLD                                     |  | 2,145  | 8.30    | 55.64  |
| ROU                                     |  | 1,257  | 4.86    | 60.51  |
| RUS                                     |  | 1,810  | 7.00    | 67.51  |
| SRB                                     |  | 1,046  | 4.05    | 71.55  |
| SVK                                     |  | 1,200  | 4.64    | 76.20  |
| TJK                                     |  | 1,200  | 4.64    | 80.84  |
| TUR                                     |  | 2,415  | 9.34    | 90.18  |
| UKR                                     |  | 1,289  | 4.99    | 95.16  |
| UZB                                     |  | 1,250  | 4.84    | 100.00 |
| -----                                   |  |        |         |        |
| Total                                   |  | 25,852 | 100.00  |        |
| -----                                   |  |        |         |        |
| -> regionname = Latin America/Caribbean |  |        |         |        |
| ISO 3166-1                              |  |        |         |        |
| alpha-3                                 |  |        |         |        |
| country                                 |  |        |         |        |
| code                                    |  | Freq.  | Percent | Cum.   |
| -----                                   |  |        |         |        |

|     |       |       |        |
|-----|-------|-------|--------|
| ARG | 1,003 | 5.75  | 5.75   |
| BOL | 2,067 | 11.85 | 17.60  |
| BRA | 1,762 | 10.10 | 27.71  |
| CHL | 1,000 | 5.73  | 33.44  |
| COL | 1,520 | 8.72  | 42.16  |
| ECU | 1,200 | 6.88  | 49.04  |
| GTM | 1,229 | 7.05  | 56.09  |
| MEX | 1,741 | 9.98  | 66.07  |
| NIC | 1,200 | 6.88  | 72.95  |
| PER | 1,400 | 8.03  | 80.98  |
| PRI | 1,127 | 6.46  | 87.44  |
| URY | 1,000 | 5.73  | 93.18  |
| VEN | 1,190 | 6.82  | 100.00 |

|       |        |        |  |
|-------|--------|--------|--|
| Total | 17,439 | 100.00 |  |
|-------|--------|--------|--|

-> regionname = Middle East/North Africa

|            |       |         |        |
|------------|-------|---------|--------|
| ISO 3166-1 |       |         |        |
| alpha-3    |       |         |        |
| country    |       |         |        |
| code       | Freq. | Percent | Cum.   |
| EGY        | 1,200 | 12.11   | 12.11  |
| IRN        | 1,499 | 15.13   | 27.25  |
| IRQ        | 1,200 | 12.11   | 39.36  |
| JOR        | 1,203 | 12.14   | 51.50  |
| LBN        | 1,200 | 12.11   | 63.62  |
| LBY        | 1,196 | 12.07   | 75.69  |
| MAR        | 1,200 | 12.11   | 87.81  |
| TUN        | 1,208 | 12.19   | 100.00 |
| Total      | 9,906 | 100.00  |        |

-> regionname = North America

|            |       |         |        |
|------------|-------|---------|--------|
| ISO 3166-1 |       |         |        |
| alpha-3    |       |         |        |
| country    |       |         |        |
| code       | Freq. | Percent | Cum.   |
| CAN        | 4,018 | 60.75   | 60.75  |
| USA        | 2,596 | 39.25   | 100.00 |
| Total      | 6,614 | 100.00  |        |

-> regionname = South Asia

|            |       |         |        |
|------------|-------|---------|--------|
| ISO 3166-1 |       |         |        |
| alpha-3    |       |         |        |
| country    |       |         |        |
| code       | Freq. | Percent | Cum.   |
| BGD        | 1,200 | 20.25   | 20.25  |
| IND        | 1,692 | 28.55   | 48.80  |
| MDV        | 1,039 | 17.53   | 66.33  |
| PAK        | 1,995 | 33.67   | 100.00 |
| Total      | 5,926 | 100.00  |        |

-> regionname = Sub-Saharan Africa

|            |       |         |        |
|------------|-------|---------|--------|
| ISO 3166-1 |       |         |        |
| alpha-3    |       |         |        |
| country    |       |         |        |
| code       | Freq. | Percent | Cum.   |
| ETH        | 1,230 | 24.86   | 24.86  |
| KEN        | 1,266 | 25.59   | 50.44  |
| NGA        | 1,237 | 25.00   | 75.44  |
| ZWE        | 1,215 | 24.56   | 100.00 |
| Total      | 4,948 | 100.00  |        |

**Table 1:** Variable definitions.

| name                                  | description                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| freedom                               | "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out." |
| place size                            | "OBSERVATIONS BY THE INTERVIEWER; Code size of town where interview was conducted"                                                                                                                                                                                                                                                                           |
| age                                   | age                                                                                                                                                                                                                                                                                                                                                          |
| male                                  | male                                                                                                                                                                                                                                                                                                                                                         |
| married or living together as married | "Are you currently ....(READ OUT AND CODE ONE ONLY) 1 'Married' 2 'Living together as married' 3 'Divorced' 4 'Separated' 5 'Widowed' 6 'Single/Never married' 7 'Divorced, Separated or Widow' 8 'Living apart but steady relation (married,cohabitation)'"                                                                                                 |
| Employment status                     | "Are you employed now or not? IF YES: About how many hours a week? If more than one job: only for the main job 1 'Full time' 2 'Part time' 3 'Self employed' 4 'Retired' 5 'Housewife' 6 'Students' 7 'Unemployed' 8 'Other'"                                                                                                                                |
| financial satisfaction                | "How satisfied are you with the financial situation of your household?"                                                                                                                                                                                                                                                                                      |
| class                                 | "Social class (subjective)"                                                                                                                                                                                                                                                                                                                                  |
| income                                | "Scale of incomes"                                                                                                                                                                                                                                                                                                                                           |
| health                                | "State of health (subjective)"                                                                                                                                                                                                                                                                                                                               |
| religion important                    | "WVS2000: For each of the following aspects, indicate how important it is in your life. Would you say it is: EVS1999: Please say, for each of the following, how important it is in your life. Religion"                                                                                                                                                     |

Per distributions it is worth noting left-skewed distribution on freedom, unexpectedly most people feel maximally free at 10 out of 1-10, likewise but as expected with urbanism—most people are in the top category  $> 500k$ , and again left-skewed on religion, probably not unexpected—religiosity is declining, but still strong.

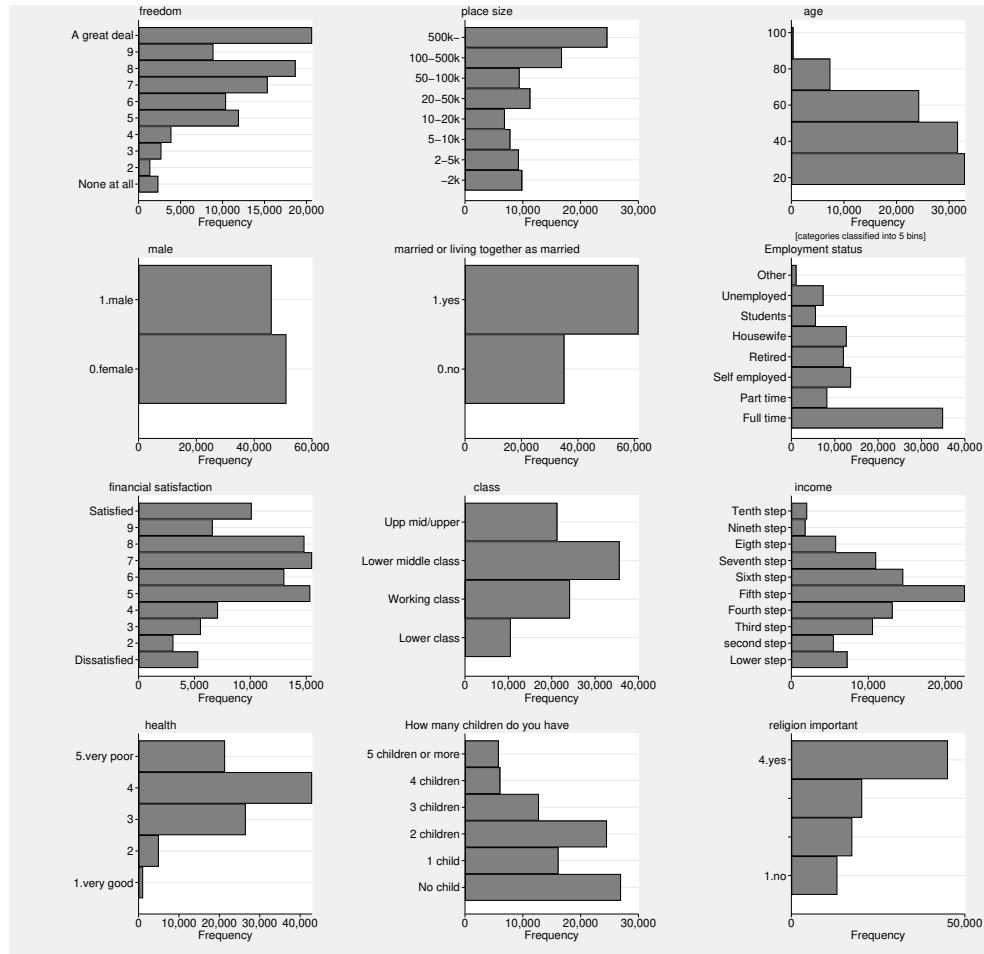


Figure 7: Variables' distribution.

## 1.2 First/initial results

### 1.2.1 by region

These were our first/initial results but they do not hold when including country dummies and so we postpone them to appendix.

First for all regions in general there is more freedom in smaller areas v top bin, the base case ( $> 500k$ ) in table 2 but controlling for other predictors in table 3 it is just the 2nd (2-5k) and 3rd (5-10k) smallest categories, and smaller values in 6th and 7th categories. But the very smallest areas ( $< 2k$ ) are not more free. Next we separate by region.

East Asia/Pacific stands out with high effect sizes at about .4 or even .5, also controlling for other predictors of freedom. Europe is mostly less free in smaller areas. Latin America/Caribbean stands out with negative effect in the very smallest areas. Middle East/North Africa has mostly negative effects. North America is positive in 2nd largest category (100-500k). South Asia and Sub-Saharan Africa mostly do not show significant differences.<sup>9</sup>

<sup>9</sup>Except .3 for South Asia for the smallest places ( $< 2k$ ).

|                | All region dummies | (with dummies) | East Asia/Pacific | Europe/Central Asia | Latin America/Caribbean | Middle East/North Africa | North America | South Asia | Sub-Saharan Africa |
|----------------|--------------------|----------------|-------------------|---------------------|-------------------------|--------------------------|---------------|------------|--------------------|
| town==2k       | -0.05+             |                | 0.05              | -0.13*              | -0.28***                | 0.22                     | -0.07         | 0.01       | -0.29              |
| town==2-5k     | 0.14***            |                | 0.36***           | 0.09                | -0.13                   | -0.09                    | -0.07         | 0.02       | 0.06               |
| town==5-10k    | 0.20***            |                | 0.51***           | 0.03                | -0.01                   | -0.12                    | 0.13          | 0.03       | 0.35               |
| town==10-20k   | 0.09**             |                | 0.52***           | -0.15**             | 0.07                    | -0.34***                 | -0.03         | -0.02      | 0.19               |
| town==20-50k   | 0.12***            |                | 0.28***           | 0.08+               | 0.08                    | -0.17*                   | 0.09          | 0.03       | -0.22              |
| town==50-100k  | 0.12***            |                | 0.41***           | -0.10+              | 0.04                    | -0.02                    | 0.12          | -0.41+     | -0.06              |
| town==100-500k | 0.08***            |                | 0.55***           | -0.12**             | -0.08+                  | -0.27***                 | 0.18**        | 0.04       | -0.45+             |
| constant       | 6.94***            |                | 6.76***           | 7.06***             | 7.85***                 | 6.88***                  | 7.44***       | 7.52***    | 7.14***            |
| N              | 93414              |                | 24560             | 25289               | 17285                   | 9846                     | 5833          | 5907       | 4694               |

+ 0.10 \* 0.05 \*\*  
0.01 \*\*\* 0.001; robust std err

Table 2: OLS regressions of freedom/agency

|                                       | All region dummies | (with dummies) | East Asia/Pacific | Europe/Central Asia | Latin America/Caribbean | Middle East/North Africa | North America | South Asia | Sub-Saharan Africa |
|---------------------------------------|--------------------|----------------|-------------------|---------------------|-------------------------|--------------------------|---------------|------------|--------------------|
| town==2k                              | -0.03              |                | -0.00             | -0.03               | -0.18**                 | 0.02                     | -0.06         | 0.28*      | -0.17              |
| town==2-5k                            | 0.09**             |                | 0.28***           | 0.00                | -0.14                   | -0.13                    | -0.10         | 0.24+      | 0.18               |
| town==5-10k                           | 0.11***            |                | 0.36***           | 0.02                | -0.03                   | -0.15+                   | -0.03         | 0.13       | 0.33               |
| town==10-20k                          | 0.03               |                | 0.41***           | -0.20***            | -0.00                   | -0.33***                 | -0.00         | 0.27       | 0.19               |
| town==20-50k                          | 0.04               |                | 0.24***           | -0.01               | -0.01                   | -0.22**                  | 0.05          | 0.02       | -0.17              |
| town==50-100k                         | 0.06*              |                | 0.34***           | -0.21***            | -0.01                   | -0.07                    | 0.13+         | 0.10       | 0.02               |
| town==100-500k                        | 0.06**             |                | 0.48***           | -0.18***            | -0.03                   | -0.30***                 | 0.13*         | 0.07       | -0.43+             |
| age                                   | 0.01***            |                | 0.01*             | 0.00                | 0.02**                  | 0.00                     | 0.03***       | 0.02       | 0.02               |
| age2                                  | -0.00***           |                | -0.00**           | -0.00               | -0.00+                  | 0.00                     | -0.00**       | -0.00      | -0.00              |
| male                                  | 0.00               |                | -0.09***          | 0.05                | -0.08*                  | 0.23***                  | -0.13**       | 0.16+      | 0.14               |
| married or living together as married | -0.01              |                | -0.00             | -0.06+              | -0.05                   | 0.01                     | -0.00         | 0.06       | 0.27*              |
| Full time                             | 0.00               |                | 0.00              | 0.00                | 0.00                    | 0.00                     | 0.00          | 0.00       | 0.00               |
| Part time                             | -0.10***           |                | -0.04             | -0.09+              | -0.24***                | -0.05                    | 0.08          | -0.32**    | -0.06              |
| Self employed                         | -0.07**            |                | -0.12**           | 0.08                | -0.04                   | -0.10                    | 0.32***       | -0.47***   | 0.15               |
| Retired                               | -0.01              |                | 0.15**            | -0.08               | 0.05                    | 0.22*                    | -0.07         | -0.31      | -0.15              |
| Housewife                             | -0.22***           |                | -0.19***          | -0.19***            | -0.05                   | -0.17*                   | -0.05         | -0.54***   | -0.15              |
| Students                              | -0.21***           |                | -0.22***          | -0.08               | -0.27***                | -0.21*                   | 0.09          | -0.31      | -0.12              |
| Unemployed                            | -0.10**            |                | -0.11+            | -0.16*              | -0.02                   | -0.14                    | -0.14         | -0.28+     | 0.12               |
| Other                                 | -0.23**            |                | -0.29+            | -0.34*              | 0.01                    | -0.20                    | -0.03         | -1.87***   | 1.53**             |
| financial satisfaction                | 0.27***            |                | 0.31***           | 0.31***             | 0.19***                 | 0.31***                  | 0.26***       | 0.31***    | 0.10***            |
| health                                | 0.34***            |                | 0.32***           | 0.25***             | 0.37***                 | 0.47***                  | 0.51***       | 0.21***    | 0.24***            |
| How many children do you have         | 0.04***            |                | -0.01             | 0.05***             | 0.02                    | 0.10***                  | 0.05**        | 0.03       | -0.10*             |
| religion important                    | 0.06***            |                | 0.09***           | 0.01                | 0.03                    | 0.37***                  | 0.08***       | 0.17**     | 0.24*              |
| constant                              | 3.55***            |                | 3.23***           | 4.26***             | 4.79***                 | 1.76***                  | 2.89***       | 3.28***    | 4.26***            |
| N                                     | 86653              |                | 23321             | 21600               | 16670                   | 9427                     | 5776          | 5231       | 4628               |

+ 0.10 \* 0.05 \*\*  
0.01 \*\*\* 0.001; robust std err

Table 3: OLS regressions of freedom/agency

## 1.2.2 Insignificant results with country dummies

Country dummies render cross-country patterns insignificant.

Here we conduct important sensitivity/robustness checks. First in a1 just urbanicity dummies. Then a1a adds region dummies like earlier in the first table little change. A1b has income level dummies TODO spell out; again little change; but addition of country dummies in a1c kills the significance and turns many categories negative notably the first smallest one, similar with controls in models a2\* and country dummies make categories even more negative in a2c than in a1c. And then a1c2 and a2c2 add clustered standard errors on country (stata: robust cluster(country))—all significance disappears except in a2c2 on the smallest areas—hence only the very smallest areas < 2k are less free than the largest ones > 500k.

in table 4 we conduct additional robustness checks, a1 and a2 are repeated from the body; a1a and a2a include region dummies: in a1a 100 – 500k results are attenuated, but others are stronger; and in a2a the top three categories spanning 20 – 500k are weaker but other categories are stronger and the smaller category < 2k positive effect goes away. But then in a1c and a2c addition of country dummies kills the effects unfortunately. And having clustered standard errors yields all results insignificant



|                                       | a1      | a1a     | a1b     | a1c      | a1c2    | a2       | a2a      | a2b      | a2c      | a2c2     |
|---------------------------------------|---------|---------|---------|----------|---------|----------|----------|----------|----------|----------|
| town==2k                              | -0.03   | -0.05+  | -0.03   | -0.07*   | -0.07   | -0.06*   | -0.03    | -0.06*   | -0.06+   | -0.06    |
| town==2-5k                            | 0.08**  | 0.14*** | 0.10**  | -0.05    | -0.05   | 0.03     | 0.09**   | 0.04     | -0.03    | -0.03    |
| town==5-10k                           | 0.16*** | 0.20*** | 0.16*** | 0.00     | 0.00    | 0.07*    | 0.11***  | 0.07*    | -0.02    | -0.02    |
| town==10-20k                          | 0.03    | 0.09**  | 0.02    | -0.12*** | -0.12   | 0.00     | 0.03     | -0.01    | -0.09**  | -0.09    |
| town==20-50k                          | 0.14*** | 0.12*** | 0.13*** | -0.02    | -0.02   | 0.07**   | 0.04     | 0.07**   | -0.04    | -0.04    |
| town==50-100k                         | 0.13*** | 0.12*** | 0.11*** | -0.01    | -0.01   | 0.10***  | 0.06*    | 0.08**   | -0.01    | -0.01    |
| town==100-500k                        | 0.13*** | 0.08*** | 0.11*** | -0.02    | -0.02   | 0.11***  | 0.06**   | 0.09***  | -0.03    | -0.03    |
| age                                   |         |         |         |          |         | 0.01+    | 0.01***  | 0.00     | 0.02***  | 0.02***  |
| age2                                  |         |         |         |          |         | -0.00*   | -0.00*** | -0.00+   | -0.00*** | -0.00*** |
| male                                  |         |         |         |          |         | 0.02     | 0.00     | 0.02     | 0.01     | 0.01     |
| married or living together as married |         |         |         |          |         | -0.08*** | -0.01    | -0.07*** | 0.00     | 0.00     |
| Full time                             |         |         |         |          |         | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     |
| Part time                             |         |         |         |          |         | -0.09*** | -0.10*** | -0.09**  | -0.09*** | -0.09*   |
| Self employed                         |         |         |         |          |         | 0.02     | -0.07**  | 0.02     | -0.02    | -0.02    |
| Retired                               |         |         |         |          |         | 0.02     | -0.01    | 0.01     | -0.05+   | -0.05    |
| Housewife                             |         |         |         |          |         | -0.16*** | -0.22*** | -0.16*** | -0.20*** | -0.20*** |
| Students                              |         |         |         |          |         | -0.19*** | -0.21*** | -0.19*** | -0.14*** | -0.14**  |
| Unemployed                            |         |         |         |          |         | -0.01    | -0.10**  | -0.00    | -0.11*** | -0.11*   |
| Other                                 |         |         |         |          |         | -0.07    | -0.23**  | -0.04    | -0.25*** | -0.25*   |
| financial satisfaction                |         |         |         |          |         | 0.27***  | 0.27***  | 0.27***  | 0.26***  | 0.26***  |
| health                                |         |         |         |          |         | 0.35***  | 0.34***  | 0.34***  | 0.34***  | 0.34***  |
| How many children do you have         |         |         |         |          |         | 0.07***  | 0.04***  | 0.07***  | 0.02*    | 0.02     |
| religion important                    |         |         |         |          |         | 0.07***  | 0.06***  | 0.07***  | 0.03***  | 0.03*    |
| region dummies                        | no      | yes     | no      | no       | no      | no       | yes      | no       | no       | no       |
| country dummies                       | no      | no      | no      | yes      | yes     | no       | no       | no       | yes      | yes      |
| income level dummies                  | no      | no      | yes     | no       | no      | no       | no       | yes      | no       | no       |
| constant                              | 7.14*** | 6.94*** | 7.14*** | 7.57***  | 7.57*** | 3.84***  | 3.55***  | 3.87***  | 3.99***  | 3.99***  |
| N                                     | 95083   | 93414   | 93414   | 95083    | 95083   | 87874    | 86653    | 86653    | 87874    | 87874    |

+ 0.10 \* 0.05 \*\*  
0.01 \*\*\* 0.001; robust std err

Table 4: OLS regressions of freedom/agency

## 1.3 Main results from paper

### 1.3.1 By country

First let's start with sample sizes by country by 3 urbanicity categories—in general they are sizeable > 100, and in cases where they are not CI is wider on the estimates such as with ETH at -1.12 on lt20k that is insignificant

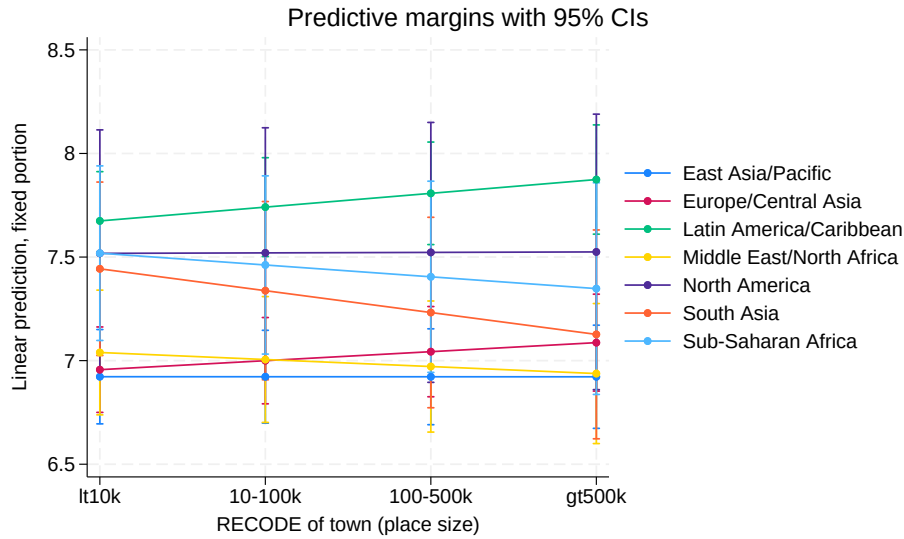
| ISO 3166-1<br>alpha-3<br>country<br>code | RECODE of town (place size) |         |        | Total |
|------------------------------------------|-----------------------------|---------|--------|-------|
|                                          | lt20k                       | 20-500k | gt500k |       |
| AND                                      | 730                         | 274     | 0      | 1,004 |
| ARG                                      | 200                         | 180     | 623    | 1,003 |
| ARM                                      | 545                         | 232     | 446    | 1,223 |
| AUS                                      | 453                         | 567     | 705    | 1,725 |
| BGD                                      | 991                         | 209     | 0      | 1,200 |
| BOL                                      | 625                         | 741     | 701    | 2,067 |
| BRA                                      | 185                         | 864     | 713    | 1,762 |
| CAN                                      | 1,098                       | 1,694   | 1,226  | 4,018 |
| CHL                                      | 152                         | 487     | 361    | 1,000 |
| CHN                                      | 50                          | 2,986   | 0      | 3,036 |
| COL                                      | 304                         | 722     | 494    | 1,520 |
| CYP                                      | 615                         | 385     | 0      | 1,000 |
| CZE                                      | 680                         | 370     | 150    | 1,200 |
| DEU                                      | 712                         | 616     | 200    | 1,528 |
| ECU                                      | 384                         | 466     | 350    | 1,200 |
| EGY                                      | 424                         | 456     | 320    | 1,200 |
| ETH                                      | 841                         | 380     | 9      | 1,230 |
| GBR                                      | 495                         | 1,713   | 397    | 2,605 |
| GRC                                      | 420                         | 340     | 440    | 1,200 |
| GTM                                      | 71                          | 505     | 653    | 1,229 |
| HKG                                      | 0                           | 0       | 2,075  | 2,075 |
| IDN                                      | 2,761                       | 346     | 93     | 3,200 |
| IND                                      | 1,415                       | 235     | 42     | 1,692 |
| IRN                                      | 400                         | 609     | 490    | 1,499 |
| IRQ                                      | 320                         | 232     | 648    | 1,200 |
| JOR                                      | 375                         | 306     | 522    | 1,203 |
| JPN                                      | 51                          | 786     | 516    | 1,353 |
| KAZ                                      | 457                         | 456     | 363    | 1,276 |
| KEN                                      | 499                         | 421     | 122    | 1,042 |
| KGZ                                      | 710                         | 290     | 200    | 1,200 |
| KOR                                      | 0                           | 364     | 881    | 1,245 |
| LBN                                      | 410                         | 660     | 130    | 1,200 |
| LBY                                      | 480                         | 544     | 172    | 1,196 |
| MAC                                      | 0                           | 0       | 1,023  | 1,023 |
| MAR                                      | 380                         | 480     | 340    | 1,200 |
| MDV                                      | 738                         | 301     | 0      | 1,039 |

|       |        |        |        |        |
|-------|--------|--------|--------|--------|
| MEX   | 858    | 463    | 403    | 1,724  |
| MMR   | 1,116  | 84     | 0      | 1,200  |
| MNG   | 378    | 257    | 1,003  | 1,638  |
| MYS   | 744    | 428    | 141    | 1,313  |
| NGA   | 848    | 375    | 14     | 1,237  |
| NIC   | 381    | 709    | 110    | 1,200  |
| NIR   | 39     | 407    | 0      | 446    |
| NLD   | 104    | 1,710  | 217    | 2,031  |
| NZL   | 420    | 387    | 187    | 994    |
| PAK   | 1,272  | 249    | 474    | 1,995  |
| PER   | 355    | 945    | 100    | 1,400  |
| PHL   | 910    | 290    | 0      | 1,200  |
| PRI   | 73     | 1,054  | 0      | 1,127  |
| ROU   | 703    | 435    | 119    | 1,257  |
| RUS   | 594    | 626    | 590    | 1,810  |
| SGP   | 0      | 0      | 2,012  | 2,012  |
| SRB   | 403    | 447    | 196    | 1,046  |
| SVK   | 750    | 450    | 0      | 1,200  |
| THA   | 1,244  | 116    | 140    | 1,500  |
| TJK   | 880    | 210    | 110    | 1,200  |
| TUN   | 514    | 614    | 80     | 1,208  |
| TUR   | 240    | 1,767  | 408    | 2,415  |
| TWN   | 40     | 1,165  | 18     | 1,223  |
| UKR   | 638    | 355    | 296    | 1,289  |
| URY   | 270    | 290    | 440    | 1,000  |
| USA   | 0      | 1,443  | 385    | 1,828  |
| UZB   | 0      | 0      | 1,250  | 1,250  |
| VEN   | 180    | 661    | 349    | 1,190  |
| VNM   | 0      | 1,049  | 151    | 1,200  |
| ZWE   | 945    | 270    | 0      | 1,215  |
| ----- |        |        |        |        |
| Total | 33,870 | 37,473 | 24,598 | 95,941 |

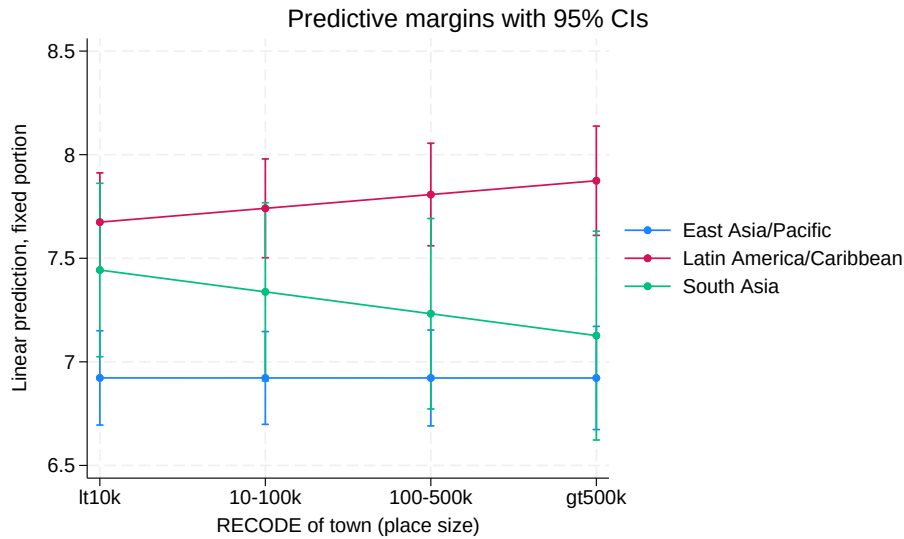
|                                             | lt20k  | 20-500k | N    |
|---------------------------------------------|--------|---------|------|
| AND                                         | -0.20+ | 0.00    | 997  |
| ARG                                         | -0.54* | -0.34+  | 849  |
| ARM                                         | 0.19   | -0.03   | 1192 |
| AUS                                         | 0.04   | -0.02   | 1575 |
| BGD                                         | 0.31*  | 0.00    | 1197 |
| BOL                                         | -0.28* | -0.25*  | 1990 |
| BRA                                         | -0.30  | -0.03   | 1687 |
| CAN                                         | 0.04   | 0.08    | 4018 |
| CHL                                         | -0.19  | -0.38*  | 903  |
| CHN                                         | -0.11  | 0.00    | 2775 |
| COL                                         | 0.06   | 0.05    | 1520 |
| CYP                                         | -0.68* | 0.00    | 943  |
| CZE                                         | -0.08  | -0.05   | 1186 |
| DEU                                         | 0.23   | 0.28+   | 1491 |
| ECU                                         | 0.04   | 0.02    | 1184 |
| EGY                                         | -0.08  | 0.03    | 1175 |
| ETH                                         | -1.12  | -1.43*  | 1215 |
| GRC                                         | 0.02   | 0.15    | 1161 |
| GTM                                         | -0.06  | -0.01   | 1113 |
| HKG                                         |        |         | 2044 |
| IDN                                         | 0.24   | 0.39    | 3154 |
| IND                                         | -0.69* | -1.12*  | 1063 |
| IRN                                         | 0.24   | 0.18    | 1477 |
| IRQ                                         | -0.06  | -0.36*  | 1177 |
| JOR                                         | 0.04   | 0.10    | 1202 |
| JPN                                         | -0.22  | -0.31*  | 1130 |
| KAZ                                         | -0.32* | -0.15   | 1116 |
| KEN                                         | -0.38  | -0.02   | 1003 |
| KGZ                                         | -0.15  | -0.05   | 1131 |
| KOR                                         |        | -0.03   | 1245 |
| LBN                                         | -0.66* | -0.39*  | 1200 |
| LBY                                         | 0.21   | -0.29   | 923  |
| MAC                                         |        |         | 792  |
| MAR                                         | 0.53*  | 0.28*   | 1200 |
| MDV                                         | 0.17   | 0.00    | 1005 |
| MEX                                         | -0.11  | -0.07   | 1619 |
| MMR                                         | -0.27  | 0.00    | 1200 |
| MNG                                         | 0.18   | 0.50*   | 1638 |
| MYS                                         | -0.32* | -0.17   | 1313 |
| NGA                                         | 0.80   | 0.16    | 1212 |
| NIC                                         | -0.42+ | -0.20   | 1200 |
| NLD                                         | 0.04   | -0.12   | 1675 |
| NZL                                         | 0.09   | 0.02    | 832  |
| PAK                                         | 0.24+  | -0.00   | 1966 |
| PER                                         | -0.65* | -0.33+  | 1387 |
| PHL                                         | -0.05  | 0.00    | 1199 |
| PRI                                         | 0.10   | 0.00    | 1065 |
| ROU                                         | 0.69*  | 0.56*   | 1088 |
| RUS                                         | -0.18  | -0.06   | 1676 |
| SGP                                         |        |         | 1966 |
| SRB                                         | -0.15  | -0.25   | 980  |
| SVK                                         | -0.10  | 0.00    | 1175 |
| THA                                         | 0.91*  | 1.29*   | 1258 |
| TJK                                         | -0.33+ | -0.28   | 1200 |
| TUN                                         | -0.08  | 0.14    | 1073 |
| TUR                                         | -0.55* | -0.39*  | 2357 |
| TWN                                         | 0.92   | 0.89    | 1221 |
| UKR                                         | -0.15  | 0.09    | 1116 |
| URY                                         | 0.02   | 0.23    | 966  |
| USA                                         |        | -0.03   | 1758 |
| UZB                                         |        |         | 1116 |
| VEN                                         | 0.09   | -0.18   | 1187 |
| VNM                                         |        | 0.22    | 1200 |
| ZWE                                         | 0.45*  | 0.00    | 1198 |
| +0.10 *0.05 **0.01 ***0.001; robust std err |        |         |      |

**Table 5:** OLS regressions of freedom/autonomy on urbanicity dummies (base: > 500k) using model controlling for age, age<sup>2</sup>, male, married, employment type dummies(full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction, health, kids, and religion important. See model for pooled sample in table 3 except that original 8 step urbanicity variable is now categorized into 3 bins as there are smaller cell sizes separately by country. Still note that few countries lack the estimates as even this simplified 3 step urbanicity spectrum is not available. Overall the results are mostly insignificant, but more negative (11) than positive (6). So in general within countries for few significant relationships there is less freedom in smaller areas as theorized in the literature.

We explore further by interacting region dummies with urban-rural in figure 8. We find significant interactions for Latin America and South Asia (v East Asia/Pacific)—it is difficult to see patters in figure with all regions—we plot only the significant interactions in figure 9.

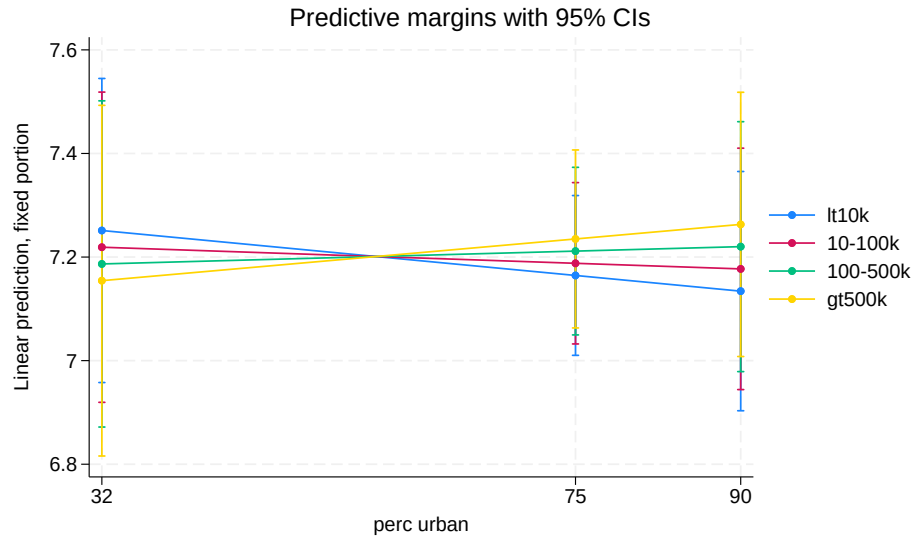


**Figure 8:** Interactions of region dummies with urban-rural.



**Figure 9:** Interactions of region dummies with urban-rural.

Last, we interacted urban-rural at person level with % urban at country level and the interaction is marginally significant at 0.1 level of significance and we plot results in figure 10. Marginsplot shows that in countries that are less urban, there is more freedom in smaller places, but in more urban countries there is more freedom in large cities. It is an intriguing finding and we do not have a good explanation for it. Perhaps in heavily urbanized countries rural barely exists and it does not help much with freedom, while in more rural countries there is more meaningful distinction between urban and rural and there is more freedom in rural areas.

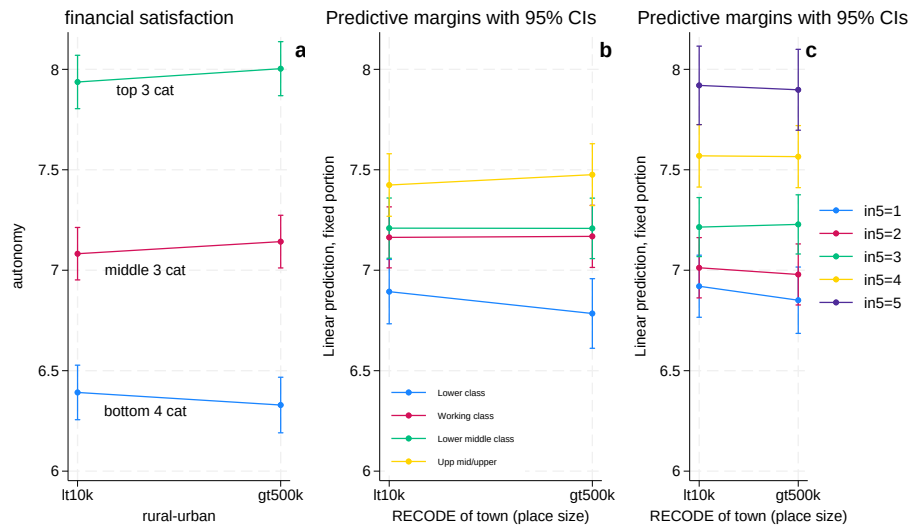


**Figure 10:** Interaction of urban-rural at person level with % urban at country level. X-axis values are chosen based on the variable's distribution: 31.2 is 5th percentile, median is 75, and 90 is 92nd percentile.

We have used above just the usual World Bank definition, which uses each country's statistical office definition—for discussion see especially Mikhaeil et al. (2024) and also Deb and Okulicz-Kozaryn (2023).

We have also tried similar cross-level interactions with PCGDP and income inequality, but didn't find significant interactions.

### 1.3.2 Interactions with financial satisfaction, class, and income



**Figure 11:** Multilevel country random intercept regressions of freedom on rural-urban dummies interacted with: a) financial satisfaction, b) class, c) income. For underlying regressions see appendix. Controls are the same as in the previous section.

|                                                        | a        | b        | c        |
|--------------------------------------------------------|----------|----------|----------|
| freedom                                                |          |          |          |
| lt10k                                                  | 0.06     | 0.11+    | 0.07     |
| 10-100k                                                | -0.07+   | 0.03     | -0.01    |
| 100-500k                                               | 0.01     | -0.05    | 0.07     |
| gt500k                                                 | 0.00     | 0.00     | 0.00     |
| RECODE of satFin (financial satisfaction)=1            | 0.00     |          |          |
| RECODE of satFin (financial satisfaction)=2            | 0.81***  |          |          |
| RECODE of satFin (financial satisfaction)=3            | 1.67***  |          |          |
| lt10k × RECODE of satFin (financial satisfaction)=1    | 0.00     |          |          |
| lt10k × RECODE of satFin (financial satisfaction)=2    | -0.12*   |          |          |
| lt10k × RECODE of satFin (financial satisfaction)=3    | -0.13*   |          |          |
| 10-100k × RECODE of satFin (financial satisfaction)=1  | 0.00     |          |          |
| 10-100k × RECODE of satFin (financial satisfaction)=2  | 0.08+    |          |          |
| 10-100k × RECODE of satFin (financial satisfaction)=3  | -0.01    |          |          |
| 100-500k × RECODE of satFin (financial satisfaction)=1 | 0.00     |          |          |
| 100-500k × RECODE of satFin (financial satisfaction)=2 | -0.04    |          |          |
| 100-500k × RECODE of satFin (financial satisfaction)=3 | -0.06    |          |          |
| gt500k × RECODE of satFin (financial satisfaction)=1   | 0.00     |          |          |
| gt500k × RECODE of satFin (financial satisfaction)=2   | 0.00     |          |          |
| gt500k × RECODE of satFin (financial satisfaction)=3   | 0.00     |          |          |
| age                                                    | 0.01***  | 0.01*    | 0.01*    |
| age2                                                   | -0.00*** | -0.00    | -0.00    |
| male                                                   | 0.01     | 0.01     | -0.00    |
| married or living together as married                  | 0.01     | 0.06**   | 0.04*    |
| Full time                                              | 0.00     | 0.00     | 0.00     |
| Part time                                              | -0.10*** | -0.11*** | -0.10*** |
| Self employed                                          | -0.02    | -0.04+   | -0.03    |
| Retired                                                | -0.06*   | -0.03    | -0.01    |
| Housewife                                              | -0.21*** | -0.20*** | -0.18*** |
| Students                                               | -0.14*** | -0.12*** | -0.11**  |
| Unemployed                                             | -0.14*** | -0.22*** | -0.18*** |
| Other                                                  | -0.27*** | -0.32*** | -0.30*** |
| health                                                 | 0.37***  | 0.50***  | 0.49***  |
| How many children do you have                          | 0.01*    | 0.01+    | 0.01+    |
| religion important                                     | 0.04***  | 0.05***  | 0.05***  |
| Lower class                                            |          | 0.00     |          |
| Working class                                          |          | 0.38***  |          |
| Lower middle class                                     |          | 0.42***  |          |
| Upp mid/upper                                          |          | 0.69***  |          |
| lt10k × Lower class                                    |          | 0.00     |          |
| lt10k × Working class                                  |          | -0.11    |          |
| lt10k × Lower middle class                             |          | -0.11    |          |
| lt10k × Upp mid/upper                                  |          | -0.16*   |          |
| 10-100k × Lower class                                  |          | 0.00     |          |
| 10-100k × Working class                                |          | -0.09    |          |
| 10-100k × Lower middle class                           |          | -0.03    |          |
| 10-100k × Upp mid/upper                                |          | -0.04    |          |
| 100-500k × Lower class                                 |          | 0.00     |          |
| 100-500k × Working class                               |          | 0.01     |          |
| 100-500k × Lower middle class                          |          | 0.03     |          |
| 100-500k × Upp mid/upper                               |          | 0.07     |          |
| gt500k × Lower class                                   |          | 0.00     |          |
| gt500k × Working class                                 |          | 0.00     |          |
| gt500k × Lower middle class                            |          | 0.00     |          |
| gt500k × Upp mid/upper                                 |          | 0.00     |          |
| RECODE of inc (income)=1                               |          |          | 0.00     |
| RECODE of inc (income)=2                               |          |          | 0.13*    |
| RECODE of inc (income)=3                               |          |          | 0.38***  |
| RECODE of inc (income)=4                               |          |          | 0.71***  |
| RECODE of inc (income)=5                               |          |          | 1.05***  |
| lt10k × RECODE of inc (income)=1                       |          |          | 0.00     |
| lt10k × RECODE of inc (income)=2                       |          |          | -0.04    |
| lt10k × RECODE of inc (income)=3                       |          |          | -0.08    |
| lt10k × RECODE of inc (income)=4                       |          |          | -0.07    |
| lt10k × RECODE of inc (income)=5                       |          |          | -0.05    |
| 10-100k × RECODE of inc (income)=1                     |          |          | 0.00     |
| 10-100k × RECODE of inc (income)=2                     |          |          | -0.01    |
| 10-100k × RECODE of inc (income)=3                     |          |          | 0.00     |
| 10-100k × RECODE of inc (income)=4                     |          |          | 0.02     |
| 10-100k × RECODE of inc (income)=5                     |          |          | -0.04    |
| 100-500k × RECODE of inc (income)=1                    |          |          | 0.00     |
| 100-500k × RECODE of inc (income)=2                    |          |          | -0.11    |
| 100-500k × RECODE of inc (income)=3                    |          |          | -0.05    |
| 100-500k × RECODE of inc (income)=4                    |          |          | -0.14    |
| 100-500k × RECODE of inc (income)=5                    |          |          | -0.27*   |
| gt500k × RECODE of inc (income)=1                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=2                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=3                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=4                      |          |          | 0.00     |
| gt500k × RECODE of inc (income)=5                      |          |          | 0.00     |
| constant                                               | 4.51***  | 4.52***  | 4.61***  |
| lns1.1.1                                               |          |          |          |
| constant                                               | -0.68*** | -0.54*** | -0.56*** |
| lnsig.e                                                |          |          |          |
| constant                                               | 0.73***  | 0.76***  | 0.76***  |
| N                                                      | 87874    | 86391    | 86471    |

+ 0.10 \* 0.05 \*\* 0.01 \*\*\* 0.001; robust std err

**Table 6:** OLS regressions of freedom/agency

We could have explored more fully here using usual sequential elaboration of the models, but for now just sticking with the final full model as there are already plenty of robustness/sensitivity checks and one could easily get lost in this sea of extra results.

## Initial interactions and alternative specifications

```
. mixed free c.town#c.satFin trust famUns criVic age age2 male mar i.emp he
> alth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184956.1
Iteration 1: Log likelihood = -184956.1
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
```

Number of obs = 86,280  
Number of groups = 64  
Obs per group:  
min = 747  
avg = 1,348.1  
max = 4,018  
Wald chi2(20) = 11068.66  
Prob > chi2 = 0.0000

Log likelihood = -184956.1

| free                | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |
|---------------------|-------------|-----------|--------|------|----------------------|
| town                | -0.03       | 0.01      | -3.99  | 0.00 | -0.05 -0.02          |
| satFin              | 0.22        | 0.01      | 31.95  | 0.00 | 0.20 0.23            |
| c.town#<br>c.satFin | 0.01        | 0.00      | 5.90   | 0.00 | 0.00 0.01            |
| trust               | 0.02        | 0.02      | 1.34   | 0.18 | -0.01 0.06           |
| famUns              | -0.11       | 0.01      | -12.26 | 0.00 | -0.13 -0.09          |
| criVic              | -0.11       | 0.03      | -4.22  | 0.00 | -0.16 -0.06          |
| age                 | 0.02        | 0.00      | 5.99   | 0.00 | 0.01 0.02            |
| age2                | -0.00       | 0.00      | -5.37  | 0.00 | -0.00 -0.00          |
| male                | 0.01        | 0.02      | 0.67   | 0.50 | -0.02 0.04           |
| mar                 | 0.00        | 0.02      | 0.03   | 0.98 | -0.03 0.03           |
| emp                 |             |           |        |      |                      |
| Part time           | -0.08       | 0.03      | -2.82  | 0.00 | -0.13 -0.02          |
| Self empl..         | -0.01       | 0.02      | -0.56  | 0.57 | -0.06 0.03           |
| Retired             | -0.04       | 0.03      | -1.32  | 0.19 | -0.10 0.02           |
| Housewife           | -0.20       | 0.03      | -8.00  | 0.00 | -0.25 -0.15          |
| Students            | -0.14       | 0.04      | -3.79  | 0.00 | -0.21 -0.07          |
| Unemployed          | -0.10       | 0.03      | -3.29  | 0.00 | -0.15 -0.04          |
| Other               | -0.22       | 0.07      | -3.28  | 0.00 | -0.36 -0.09          |
| health              | 0.33        | 0.01      | 37.13  | 0.00 | 0.31 0.35            |
| kids                | 0.02        | 0.01      | 2.67   | 0.01 | 0.00 0.03            |
| rel_imp             | 0.04        | 0.01      | 4.11   | 0.00 | 0.02 0.05            |
| _cons               | 4.26        | 0.11      | 38.13  | 0.00 | 4.04 4.48            |

| Random-effects parameters | Estimate | Std. err. | [95% conf. interval] |
|---------------------------|----------|-----------|----------------------|
| cc: Identity              |          |           |                      |
| var(_cons)                | 0.28     | 0.05      | 0.19 0.39            |
| var(Residual)             | 4.25     | 0.02      | 4.21 4.29            |

LR test vs. linear model: chibar2(01) = 4378.78 Prob >= chibar2 = 0.0000

.19

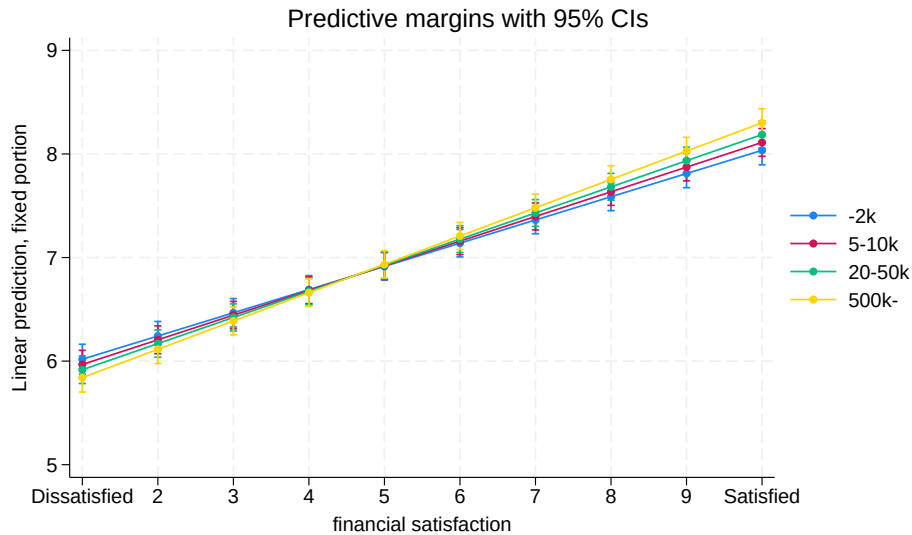


Figure 12: ini1

few things to note in the above—we have trust, famUns and criVic—which are rather predicted by urbanicity, not predictors, hence we may have overcontrol bias. Also here both town and satFin are forced to be linear, a tenuous assumption.

```

mixed free i.t4##i.sf3 trust famUns criVic age age2 male mar i.emp health
> kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -185241.39
Iteration 1: Log likelihood = -185241.39 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc

```

```

Number of obs   = 86,280
Number of groups = 64
Obs per group:
    min = 747
    avg = 1,348.1
    max = 4,018
Wald chi2(28)   = 10427.08
Prob > chi2     = 0.0000

```

Log likelihood = -185241.39

| free        | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |       |
|-------------|-------------|-----------|--------|------|----------------------|-------|
| t4          |             |           |        |      |                      |       |
| 10-100k     | -0.14       | 0.04      | -3.39  | 0.00 | -0.21                | -0.06 |
| 100-500k    | -0.04       | 0.05      | -0.78  | 0.43 | -0.13                | 0.06  |
| gt500k      | -0.03       | 0.04      | -0.74  | 0.46 | -0.12                | 0.05  |
| sf3         |             |           |        |      |                      |       |
| 2           | 0.65        | 0.03      | 18.99  | 0.00 | 0.59                 | 0.72  |
| 3           | 1.50        | 0.04      | 40.61  | 0.00 | 1.42                 | 1.57  |
| t4#sf3      |             |           |        |      |                      |       |
| 10-100k#2   | 0.21        | 0.05      | 4.38   | 0.00 | 0.12                 | 0.31  |
| 10-100k#3   | 0.12        | 0.05      | 2.38   | 0.02 | 0.02                 | 0.22  |
| 100-500k#2  | 0.09        | 0.06      | 1.62   | 0.11 | -0.02                | 0.20  |
| 100-500k#3  | 0.07        | 0.06      | 1.17   | 0.24 | -0.05                | 0.19  |
| gt500k#2    | 0.12        | 0.05      | 2.43   | 0.01 | 0.02                 | 0.22  |
| gt500k#3    | 0.12        | 0.05      | 2.26   | 0.02 | 0.02                 | 0.23  |
| trust       | 0.03        | 0.02      | 1.80   | 0.07 | -0.00                | 0.07  |
| famUns      | -0.12       | 0.01      | -13.44 | 0.00 | -0.14                | -0.10 |
| criVic      | -0.12       | 0.03      | -4.75  | 0.00 | -0.17                | -0.07 |
| age         | 0.02        | 0.00      | 5.32   | 0.00 | 0.01                 | 0.02  |
| age2        | -0.00       | 0.00      | -4.66  | 0.00 | -0.00                | -0.00 |
| male        | 0.01        | 0.02      | 0.70   | 0.48 | -0.02                | 0.04  |
| mar         | 0.01        | 0.02      | 0.34   | 0.73 | -0.03                | 0.04  |
| emp         |             |           |        |      |                      |       |
| Part time   | -0.08       | 0.03      | -3.06  | 0.00 | -0.14                | -0.03 |
| Self empl.. | -0.02       | 0.02      | -0.70  | 0.48 | -0.06                | 0.03  |
| Retired     | -0.05       | 0.03      | -1.66  | 0.10 | -0.11                | 0.01  |
| Housewife   | -0.22       | 0.03      | -8.41  | 0.00 | -0.27                | -0.17 |
| Students    | -0.14       | 0.04      | -3.88  | 0.00 | -0.21                | -0.07 |
| Unemployed  | -0.13       | 0.03      | -4.41  | 0.00 | -0.19                | -0.07 |
| Other       | -0.25       | 0.07      | -3.65  | 0.00 | -0.39                | -0.12 |
| health      | 0.35        | 0.01      | 39.72  | 0.00 | 0.34                 | 0.37  |
| kids        | 0.01        | 0.01      | 2.26   | 0.02 | 0.00                 | 0.03  |
| rel_imp     | 0.04        | 0.01      | 4.41   | 0.00 | 0.02                 | 0.06  |
| _cons       | 4.82        | 0.11      | 45.42  | 0.00 | 4.61                 | 5.03  |

| Random-effects parameters | Estimate | Std. err. | [95% conf. interval] |      |
|---------------------------|----------|-----------|----------------------|------|
| cc: Identity              |          |           |                      |      |
| var(_cons)                | 0.27     | 0.05      | 0.19                 | 0.39 |
| var(Residual)             | 4.28     | 0.02      | 4.23                 | 4.32 |

LR test vs. linear model: chibar2(01) = 4250.93      Prob >= chibar2 = 0.0000



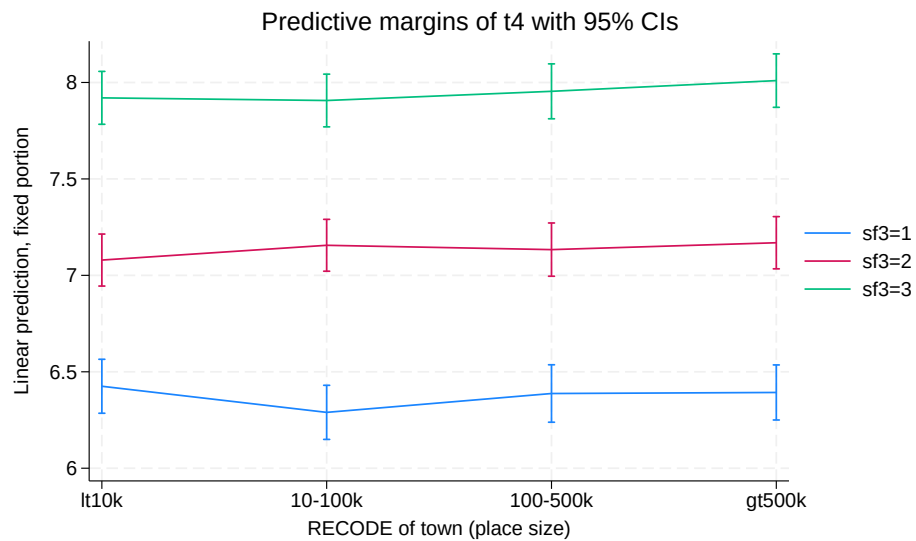


Figure 13: ini2

In the above a different take—financial satisfaction tertiles—for the 1st tertile the biggest drop is actually in the middle category at 10-100k.

```

mixed free c.town#c.inc trust famUns criVic age age2 male mar i.emp healt
> h kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184725.92
Iteration 1: Log likelihood = -184725.92 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs = 84,979
Number of groups = 64
Obs per group:
    min = 749
    avg = 1,327.8
    max = 4,018
Wald chi2(20) = 5460.49
Prob > chi2 = 0.0000
Log likelihood = -184725.92

```

|              | free | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |       |
|--------------|------|-------------|-----------|--------|------|----------------------|-------|
| town         |      | -0.00       | 0.01      | -0.43  | 0.67 | -0.02                | 0.01  |
| inc          |      | 0.11        | 0.01      | 13.01  | 0.00 | 0.09                 | 0.12  |
| c.town#c.inc |      | 0.00        | 0.00      | 1.21   | 0.23 | -0.00                | 0.00  |
| trust        |      | 0.07        | 0.02      | 3.91   | 0.00 | 0.04                 | 0.11  |
| famUns       |      | -0.17       | 0.01      | -18.82 | 0.00 | -0.19                | -0.16 |
| criVic       |      | -0.16       | 0.03      | -6.05  | 0.00 | -0.21                | -0.11 |
| age          |      | 0.01        | 0.00      | 2.60   | 0.01 | 0.00                 | 0.01  |
| age2         |      | -0.00       | 0.00      | -1.15  | 0.25 | -0.00                | 0.00  |
| male         |      | -0.00       | 0.02      | -0.17  | 0.87 | -0.03                | 0.03  |
| mar          |      | 0.04        | 0.02      | 1.97   | 0.05 | 0.00                 | 0.07  |
| emp          |      |             |           |        |      |                      |       |
| Part time    |      | -0.08       | 0.03      | -2.83  | 0.00 | -0.14                | -0.02 |
| Self empl..  |      | -0.01       | 0.02      | -0.46  | 0.65 | -0.06                | 0.04  |
| Retired      |      | 0.01        | 0.03      | 0.33   | 0.74 | -0.05                | 0.07  |
| Housewife    |      | -0.18       | 0.03      | -6.65  | 0.00 | -0.23                | -0.12 |
| Students     |      | -0.11       | 0.04      | -2.99  | 0.00 | -0.18                | -0.04 |
| Unemployed   |      | -0.15       | 0.03      | -5.04  | 0.00 | -0.21                | -0.09 |
| Other        |      | -0.25       | 0.07      | -3.54  | 0.00 | -0.39                | -0.11 |
| health       |      | 0.46        | 0.01      | 51.21  | 0.00 | 0.45                 | 0.48  |
| kids         |      | 0.01        | 0.01      | 1.93   | 0.05 | -0.00                | 0.03  |
| rel_imp      |      | 0.05        | 0.01      | 5.88   | 0.00 | 0.04                 | 0.07  |
| _cons        |      | 4.75        | 0.12      | 39.88  | 0.00 | 4.51                 | 4.98  |

| Random-effects parameters |               | Estimate | Std. err. | [95% conf. interval] |      |
|---------------------------|---------------|----------|-----------|----------------------|------|
| cc: Identity              |               |          |           |                      |      |
|                           | var(_cons)    | 0.35     | 0.06      | 0.25                 | 0.50 |
|                           | var(Residual) | 4.51     | 0.02      | 4.47                 | 4.55 |

LR test vs. linear model: chibar2(01) = 5398.66 Prob >= chibar2 = 0.0000

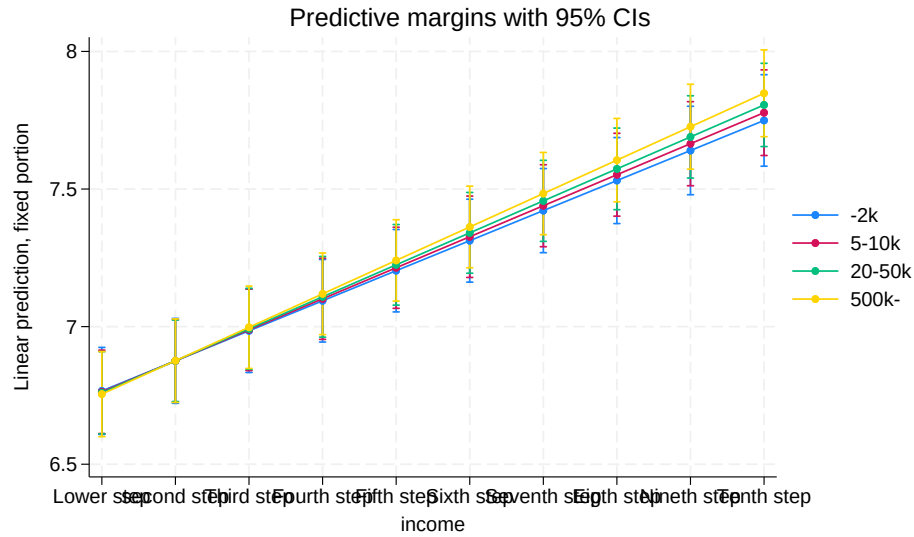


Figure 14: ini3

Here what is easy to see is that along the income spectrum the largest places  $> 500k$  change most; and smallest places  $< 2k$  change the least

```

mixed free c.town##c.class trust famUns criVic age age2 male mar i.emp hea
> lth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184877.86
Iteration 1: Log likelihood = -184877.86
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc

Number of obs   = 84,931
Number of groups = 64
Obs per group:
    min = 686
    avg = 1,327.0
    max = 4,018
Wald chi2(20)   = 4850.01
Prob > chi2     = 0.0000
Log likelihood = -184877.86

```

|             | free | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |       |
|-------------|------|-------------|-----------|--------|------|----------------------|-------|
| town        |      | -0.00       | 0.01      | -0.52  | 0.60 | -0.02                | 0.01  |
| class       |      | 0.14        | 0.02      | 7.62   | 0.00 | 0.10                 | 0.18  |
| c.town#     |      |             |           |        |      |                      |       |
| c.class     |      | 0.01        | 0.00      | 1.56   | 0.12 | -0.00                | 0.01  |
| trust       |      | 0.09        | 0.02      | 4.79   | 0.00 | 0.05                 | 0.13  |
| famUns      |      | -0.18       | 0.01      | -19.22 | 0.00 | -0.20                | -0.16 |
| criVic      |      | -0.17       | 0.03      | -6.34  | 0.00 | -0.22                | -0.11 |
| age         |      | 0.01        | 0.00      | 2.45   | 0.01 | 0.00                 | 0.01  |
| age2        |      | -0.00       | 0.00      | -1.24  | 0.22 | -0.00                | 0.00  |
| male        |      | 0.01        | 0.02      | 0.41   | 0.68 | -0.02                | 0.04  |
| mar         |      | 0.05        | 0.02      | 2.96   | 0.00 | 0.02                 | 0.09  |
| emp         |      |             |           |        |      |                      |       |
| Part time   |      | -0.10       | 0.03      | -3.45  | 0.00 | -0.16                | -0.04 |
| Self empl.. |      | -0.03       | 0.02      | -1.22  | 0.22 | -0.08                | 0.02  |
| Retired     |      | -0.02       | 0.03      | -0.64  | 0.52 | -0.08                | 0.04  |
| Housewife   |      | -0.21       | 0.03      | -7.84  | 0.00 | -0.26                | -0.16 |
| Students    |      | -0.13       | 0.04      | -3.46  | 0.00 | -0.20                | -0.06 |
| Unemployed  |      | -0.21       | 0.03      | -6.93  | 0.00 | -0.27                | -0.15 |
| Other       |      | -0.30       | 0.07      | -4.13  | 0.00 | -0.44                | -0.16 |
| health      |      | 0.48        | 0.01      | 52.29  | 0.00 | 0.46                 | 0.49  |
| kids        |      | 0.01        | 0.01      | 1.81   | 0.07 | -0.00                | 0.02  |
| rel_imp     |      | 0.05        | 0.01      | 5.88   | 0.00 | 0.04                 | 0.07  |
| _cons       |      | 4.86        | 0.12      | 39.41  | 0.00 | 4.61                 | 5.10  |

| Random-effects parameters |               | Estimate | Std. err. | [95% conf. interval] |      |
|---------------------------|---------------|----------|-----------|----------------------|------|
| cc: Identity              |               |          |           |                      |      |
|                           | var(_cons)    | 0.36     | 0.06      | 0.25                 | 0.51 |
|                           | var(Residual) | 4.54     | 0.02      | 4.49                 | 4.58 |

LR test vs. linear model: chibar2(01) = 5460.71 Prob >= chibar2 = 0.0000

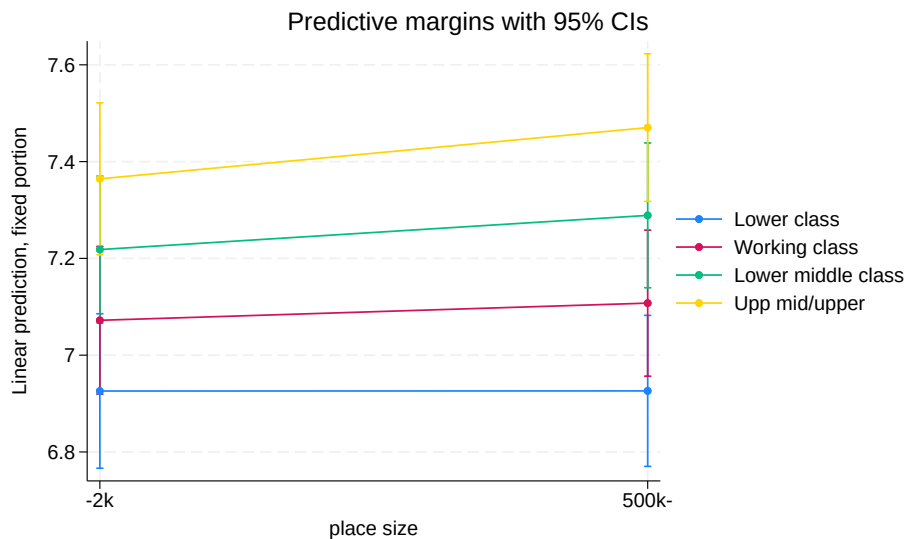


Figure 15: ini4

The above is a version of the main graph from the body of the paper, again here we are over-controlling with trust and the 2 crime variables, but it is not that different after all, similar patters, just the main difference being that here the lower class is flat and in the body downward sloping and here the top class is quite steep and while in paper it is upward sloping it's little less so in the body of the paper

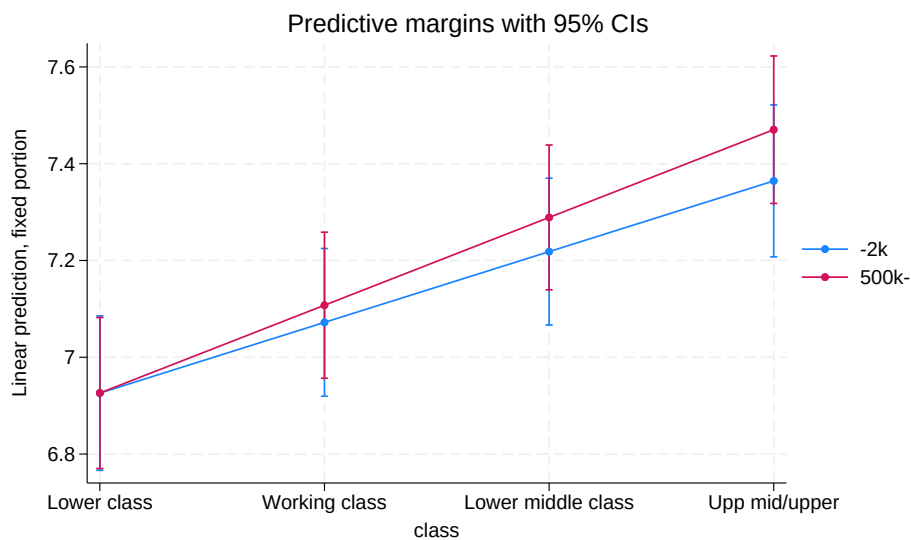


Figure 16: ini5

ini5 is from the same model, nicely shows that the higher the class the more difference between the smallest and the largest places

```

mixed free c.town##i.class trust famUns criVic age age2 male mar i.emp hea
> lth kids rel_imp || cc: ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0: Log likelihood = -184853.93
Iteration 1: Log likelihood = -184853.93 (backed up)
Computing standard errors ...
Mixed-effects ML regression
Group variable: cc
Number of obs   = 84,931
Number of groups = 64
Obs per group:
    min = 686
    avg = 1,327.0
    max = 4,018

```

Log likelihood = -184853.93      Wald chi2(24) = 4900.64  
 Prob > chi2 = 0.0000

| free         | Coefficient | Std. err. | z      | P> z | [95% conf. interval] |
|--------------|-------------|-----------|--------|------|----------------------|
| town         | -0.00       | 0.01      | -0.54  | 0.59 | -0.02 0.01           |
| class        |             |           |        |      |                      |
| Working c..  | 0.22        | 0.06      | 3.86   | 0.00 | 0.11 0.34            |
| Lower mid..  | 0.28        | 0.06      | 5.10   | 0.00 | 0.17 0.39            |
| Upp mid/u..  | 0.48        | 0.06      | 7.75   | 0.00 | 0.36 0.60            |
| class#c.town |             |           |        |      |                      |
| Working c..  | 0.01        | 0.01      | 1.34   | 0.18 | -0.01 0.03           |
| Lower mid..  | 0.01        | 0.01      | 1.28   | 0.20 | -0.01 0.03           |
| Upp mid/u..  | 0.02        | 0.01      | 1.95   | 0.05 | -0.00 0.04           |
| trust        | 0.09        | 0.02      | 4.82   | 0.00 | 0.05 0.13            |
| famUns       | -0.18       | 0.01      | -19.18 | 0.00 | -0.20 -0.16          |
| criVic       | -0.16       | 0.03      | -6.30  | 0.00 | -0.22 -0.11          |
| age          | 0.01        | 0.00      | 2.50   | 0.01 | 0.00 0.01            |
| age2         | -0.00       | 0.00      | -1.29  | 0.20 | -0.00 0.00           |
| male         | 0.01        | 0.02      | 0.35   | 0.72 | -0.03 0.04           |
| mar          | 0.05        | 0.02      | 2.87   | 0.00 | 0.02 0.09            |
| emp          |             |           |        |      |                      |
| Part time    | -0.10       | 0.03      | -3.39  | 0.00 | -0.15 -0.04          |
| Self empl..  | -0.03       | 0.02      | -1.10  | 0.27 | -0.07 0.02           |
| Retired      | -0.02       | 0.03      | -0.52  | 0.60 | -0.08 0.05           |
| Housewife    | -0.20       | 0.03      | -7.60  | 0.00 | -0.25 -0.15          |
| Students     | -0.13       | 0.04      | -3.34  | 0.00 | -0.20 -0.05          |
| Unemployed   | -0.20       | 0.03      | -6.63  | 0.00 | -0.26 -0.14          |
| Other        | -0.29       | 0.07      | -4.01  | 0.00 | -0.43 -0.15          |
| health       | 0.47        | 0.01      | 52.12  | 0.00 | 0.46 0.49            |
| kids         | 0.01        | 0.01      | 1.85   | 0.06 | -0.00 0.03           |
| rel_imp      | 0.05        | 0.01      | 5.85   | 0.00 | 0.04 0.07            |
| _cons        | 4.96        | 0.12      | 40.78  | 0.00 | 4.72 5.20            |

19

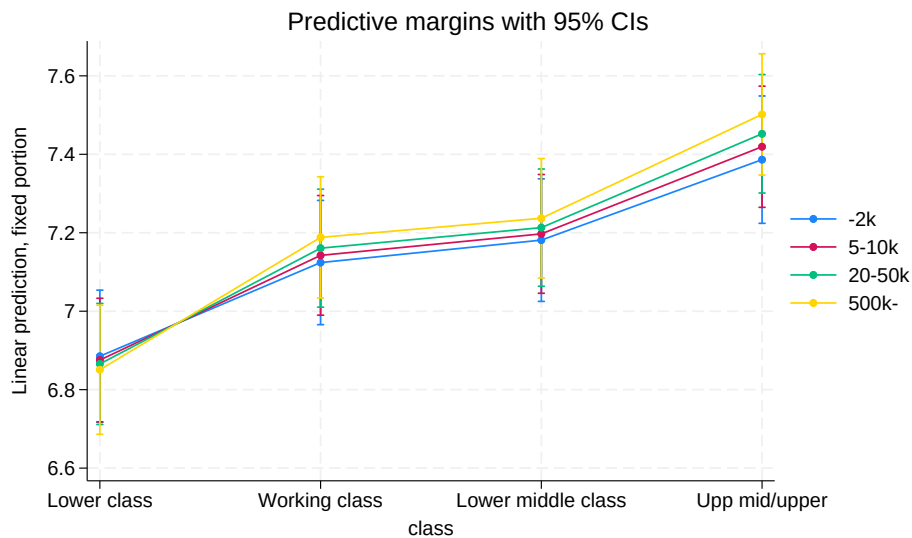


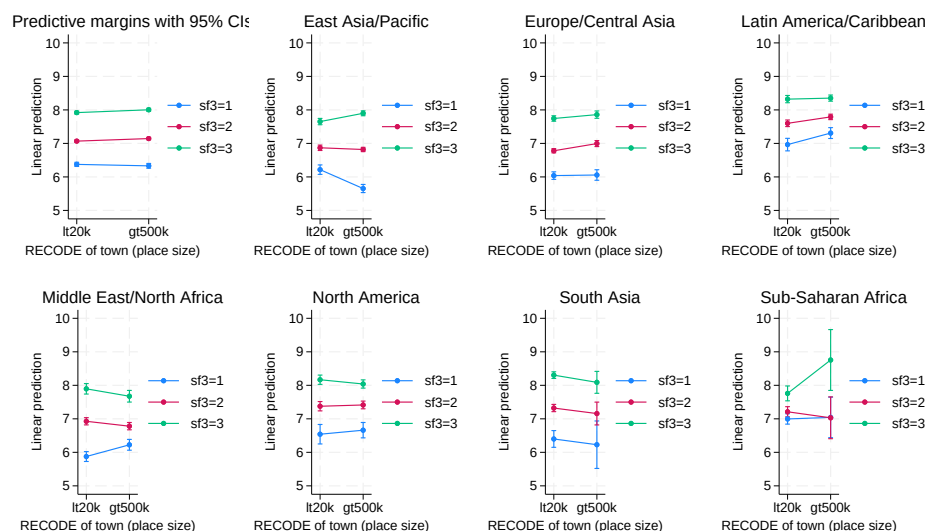
Figure 17: ini6

And then ini6: couple interesting things: first things flip from lower to working class, then working and lower middle class quite similar, and the differences get larger for upper middle and upper classes.

### 1.3.3 By region and country income group: Interactions with financial satisfaction

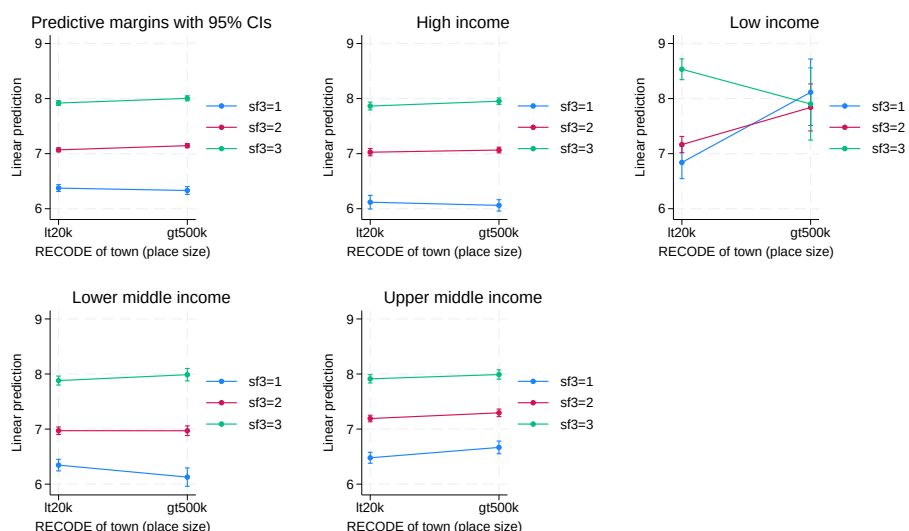
Here we repeat urbanicity interactions with financial satisfaction from the body of the paper but by region and income group; the first panel in each graph is repeated from the body of the paper.

In figure 18 we see that the pattern is much stronger for East Asia/Pacific, and opposite especially in Middle East/North Africa, but also in North America.



**Figure 18:** Urbanicity interactions with financial satisfaction by region.

In figure 19 high income has a similar pattern, and lower middle income panel has a strong pattern, and quite the opposite pattern for upper middle income.



**Figure 19:** Urbanicity interactions with financial satisfaction by income group. Note just 2 countries for low income group: ETH TJK.

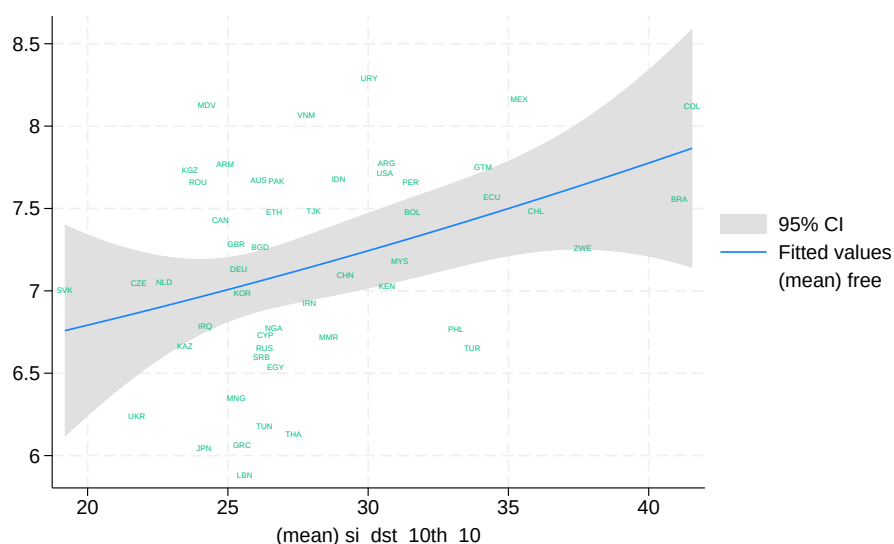
## 2 Auxiliary results and sidenote

### 2.1 Some graphs

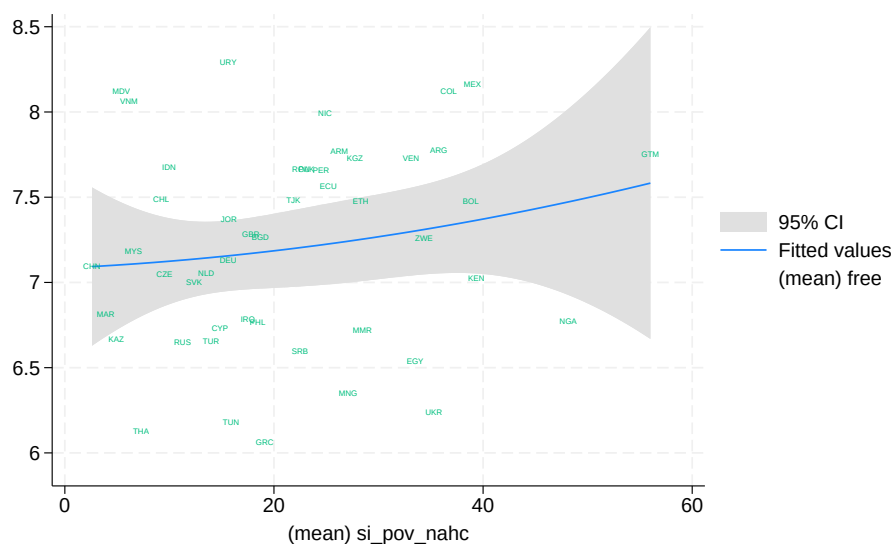
fig 20 and 21

weird to think that more ineq and pov would produce more freedom/autonomy; rather i guess lax capitalist friendly regulation would produce more ineq and pov, but should also produce less autonomy/freedom on avg, but it doesn't, there is positive relationship

or maybe the other way round: more freedom/autonomy more ineq and pov



**Figure 20:** more ineq (inc share held by top 10perc, more autonomy—conservatives would like it!! and pretty strong corr .4



**Figure 21:** likewise with poverty, but here corr just .2

## 2.2 Literature

Newman (2020) makes some useful points that may be worth citing in full:

Social theory typically conceives of urban growth as associated with political freedom. [...] But not even in the most likely of cases did the city makes urban peasants free. Focusing on Mexico City, I argue that because rural-to-urban migrants became urban denizens by establishing squatter settlements, they entered into dependent relations with political elites, which reinforced a system of political dependency, or clientelist politics. [...] Mexico is, perhaps, the most iconic case. A large cross section of the peasantry rose in arms against landowning oligarchs in the insurrectionary phase of the Mexican Revolution (1910-1917), ultimately precipitating the wholesale reconstruction of the political order. Thereafter, peasants continued to mobilize for radical change, constituting a major obstacle to postrevolutionary governing elites who sought to consolidate control. [...] When a large and growing portion of rural-to-urban migrants live in squatter settlements, and

therefore have little recourse to legal protection in residential matters, they almost inevitably grow dependent on political elites—in an arrangement known as “clientelism.” Urban concentration gives rise to clientelism.

```
. tabstat ls, by(lwd7) stat(mean sd) format(%9.1f)
```

Summary for variables: ls  
Group variable: lwd7 (RECODE of lwd (do what you like))

| lwd7  | Mean | SD  |
|-------|------|-----|
| 0-4   | 5.6  | 2.7 |
| 2     | 7.6  | 2.3 |
| 3     | 7.6  | 1.8 |
| 4     | 8.0  | 1.5 |
| 5     | 7.8  | 1.9 |
| 6     | 8.2  | 1.5 |
| 7     | 8.9  | 1.5 |
| Total | 8.3  | 1.9 |

.9

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